

AIDS 2025 May PYQ Answers

Q1. [5 Marks each]

- a) What are the different issues in ML algorithms?
- b) Explain PEAS descriptors. Provide PEAS descriptors for Part Picking robot.
- c) Define and state effects of overfitting and underfitting.
- d) Explain goal-based agents with example.

Q2. [10 Marks each]

- a) Differentiate between data scientists, big data professionals and data analysts.
- b) Explain state space based problem formulation. Formulate 8 Puzzle and N Queens problem.

Q3. [10 Marks each]

- a) Compare Informed and Uninformed Search algorithms. Explain working of A* algorithm with an example. Explain which category of search algorithms it belongs to.
- b) What are the different univariate plots in EDA? Explain them in detail.

Q4. [10 Marks each]

- a) Consider following data. Draw scatter plot for this data. Comment on the relationship between Observed Value 1 and Observed Value 2 based on the graph. Find Coefficient of correlation. Does this statistic confirm your observation? Explain your answer.

Experiment No.	1	2	3	4	5	6	7	8	9	10
Observed Value 1	20	10	30	15	45	40	30	35	45	30
Observed value 2	20	15	30	10	40	65	20	35	90	55

b) Explain the working of the Hill climbing algorithm. What are issues in Hill climbing?

Q5. [10 Marks each]

- a) Describe steps for developing ML applications with a labeled diagram.
- b) What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.

Q6. [5 Marks each]

Write Short Note on (Any four):

- a) ANOVA
- b) Min Max Algorithm
- c) Non-graphical EDA
- d) Forward chaining based proofs
- e) Box Plot

Q1. [5 Marks each] - Answers

a. What are the different issues in ML algorithms?

Definition:

Machine Learning algorithms face several challenges that affect their accuracy, efficiency, and reliability.

Key Issues:

- **Overfitting:** Model learns noise in training data, performs poorly on unseen data.
- **Underfitting:** Model is too simple, fails to capture important patterns.
- **Bias–Variance Tradeoff:** Balancing model complexity to avoid high bias or high variance.

- **Data Quality Problems:** Missing values, noisy or imbalanced datasets reduce performance.
- **Computational Complexity:** Large datasets and complex models require high time and resources.
- **Scalability:** Difficulty in handling big data efficiently.

Example:

- In spam detection, if the model is **overfitted**, it may classify unusual but harmless emails as spam.
- If **underfitted**, it may fail to detect actual spam messages.

b. Explain PEAS descriptors. Provide PEAS descriptors for Part Picking robot.

Definition:

PEAS stands for **Performance measure, Environment, Actuators, Sensors**. It is a framework used to describe the components of an intelligent agent.

Explanation:

- **Performance Measure:** Criteria to evaluate success of the agent.
- **Environment:** External conditions in which the agent operates.
- **Actuators:** Mechanisms through which the agent acts on the environment.
- **Sensors:** Tools to perceive and gather information from the environment.

PEAS for Part-Picking Robot

- **Performance Measure:** Accuracy in picking correct parts, speed, minimal damage/errors.
- **Environment:** Factory floor, conveyor belts, bins, shelves, other robots/workers.
- **Actuators:** Robotic arm, grippers, wheels (if mobile).
- **Sensors:** Cameras, proximity sensors, RFID/barcode scanners, force sensors.

c. Define and state effects of overfitting and underfitting.

Definition:

- **Overfitting:** Model learns training data too closely, including noise, leading to poor generalization.
- **Underfitting:** Model is too simple, fails to capture patterns, resulting in poor accuracy.

Effects of Overfitting:

- High accuracy on training data but low accuracy on test data.
- Model memorizes noise and irrelevant details.
- Poor generalization to unseen datasets.
- Misleadingly optimistic training results.
- Needs regularization, pruning, or more data to fix.

Effects of Underfitting:

- Low accuracy on both training and test data.
- Model ignores important relationships in data.
- Predictions are overly simplistic.
- High bias, low variance.
- Improved by using complex models or better features.

d. Explain goal-based agents with example.

Definition:

A **Goal-Based Agent** is an AI agent that selects actions by considering future consequences and choosing those that help achieve a defined goal.

Key Points:

- Goes beyond simple condition–action rules.

- Uses **goal information** to decide actions.
- Employs **search and planning** to find action sequences.
- Considers **future states** before acting.
- More flexible than reflex agents.
- Can adapt to different environments and goals.

Example:

- **Autonomous Taxi:** The agent's goal is to safely deliver passengers to their destination. It plans routes, adapts to traffic, and chooses actions (turns, stops) that move toward the goal.

Q2. [10 Marks each] - Answers

a. Differentiate between data scientists, big data professionals and data analysts.

Aspect	Data Scientist	Big Data Professional	Data Analyst
Definition	Expert who uses advanced techniques to extract insights and build predictive models	Expert who manages and processes large-scale data systems	Professional who analyzes structured data to generate reports and insights
Focus Area	Prediction, machine learning, AI models	Handling massive datasets (Big Data infrastructure)	Data interpretation and reporting
Type of Data	Structured + Unstructured	Very large-scale structured & unstructured data	Mostly structured data
Tools & Technologies	Python, R, TensorFlow, Scikit-learn	Hadoop, Spark, Kafka, NoSQL	Excel, SQL, Power BI, Tableau
Skills Required	Statistics, ML, programming, data modeling	Distributed computing, data engineering, system design	Data visualization, querying, basic statistics

Aspect	Data Scientist	Big Data Professional	Data Analyst
Main Tasks	Build models, predictive analysis, data mining	Data storage, processing, pipeline building	Data cleaning, reporting, dashboard creation
Complexity Level	High	High (technical infrastructure)	Moderate
Output	Predictive models, intelligent systems	Processed and scalable data systems	Reports, charts, dashboards
Example	Predicting customer churn using ML	Managing petabytes of data in Hadoop cluster	Creating sales reports using Excel/Tableau

b. Explain state space based problem formulation. Formulate 8 Puzzle and N Queens problem.

Definition:

State space formulation represents a problem as a set of states, actions, and transitions. It defines how an agent moves from an initial state to a goal state through valid operations.

Main Explanation

- **State Space:** All possible states reachable from the initial state.
- **Initial State:** Starting point of the problem.
- **Goal State:** Desired solution state.
- **Operators (Actions):** Rules that transform one state into another.
- **Path Cost:** Numerical cost of moving through states (if relevant).
- **Solution:** Sequence of actions leading from initial to goal state.

Formulation Examples

1. 8 Puzzle Problem

- **Initial State:** A 3×3 board with tiles numbered 1–8 and one blank space.
- **State Space:** All possible arrangements of tiles (9! states).

- **Operators:** Move blank tile up, down, left, or right.
- **Goal State:** Tiles arranged in order (1–8) with blank at bottom-right.
- **Path Cost:** Each move costs 1.
- **Example:**

2	8	3
1	6	4
7		5

Initial State

1	2	3
8		4
7	6	5

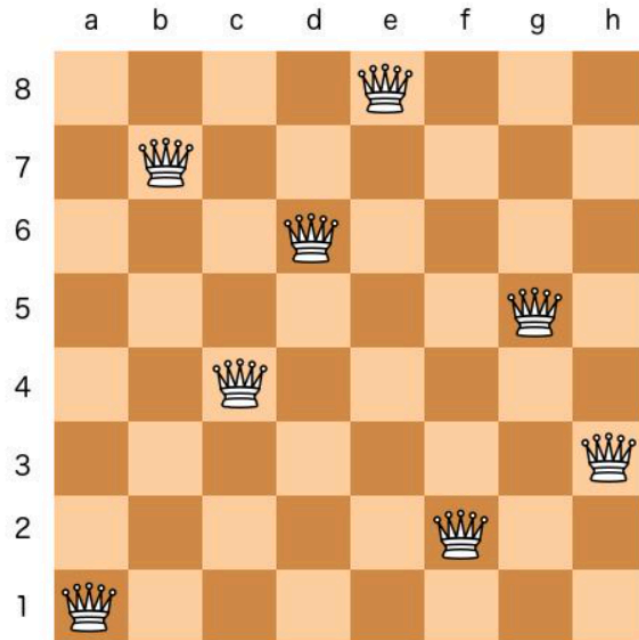
Goal State A

	1	2
3	4	5
6	7	8

Goal State B

2. N Queens Problem

- **Initial State:** Empty $N \times N$ chessboard.
- **State Space:** All possible placements of N queens on the board.
- **Operators:** Place a queen in a row/column ensuring no attack.
- **Goal State:** N queens placed such that none share the same row, column, or diagonal.
- **Path Cost:** Not relevant (any valid arrangement is acceptable).
- **Example (for 8 Queens):** One queen per row and column, avoiding diagonals.



Q3. [10 Marks each] - Answers

a. Compare Informed and Uninformed Search algorithms. Explain working of A* algorithm with an example. Explain which category of search algorithms it belongs to.

Aspect	Uninformed Search	Informed Search
Definition	Uses no extra knowledge beyond the problem definition	Uses heuristic knowledge to guide the search
Knowledge Used	Only initial state, operators, goal test	Uses heuristic function ($h(n)$)
Search Strategy	Explores blindly	Explores intelligently toward goal
Efficiency	Less efficient (more time & memory)	More efficient (faster solution)
Examples	BFS, DFS, Uniform Cost Search	A*, Greedy Best First Search
Node Expansion	Expands many unnecessary nodes	Expands fewer relevant nodes

Aspect	Uninformed Search	Informed Search
Optimality	Some are optimal (e.g., BFS)	Can be optimal (e.g., A*)
Complexity	High	Reduced using heuristics

Working of A Algorithm

A* is a best-first search algorithm that selects the node with the lowest evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$: Cost from start to current node
- $h(n)$: Heuristic (estimated cost to goal)
- $f(n)$: Total estimated cost

Steps of A*:

1. Start from the initial node and add it to the **open list**.
2. Select node with lowest $f(n)$.
3. If it is the goal → stop.
4. Otherwise, expand the node and generate successors.
5. Compute $f(n)$ for each successor.
6. Add successors to open list and repeat.

Example of A*

Consider a graph:

- Start node: **S**
- Goal node: **G**

Node	$g(n)$	$h(n)$	$f(n)$
S	0	7	7
A	1	6	7

Node	$g(n)$	$h(n)$	$f(n)$
B	4	2	6
G	6	0	6

Working:

- Start at S → expand neighbors A and B
- Compare $f(A)=7$, $f(B)=6$ → choose B
- From B → reach G
- Path found: **S** → **B** → **G** (optimal path)

Category of A*

A* belongs to **Informed Search Algorithms** because:

- It uses heuristic function $h(n)$
- It combines actual cost and estimated cost
- It guides search efficiently toward goal

b. What are the different univariate plots in EDA? Explain them in detail.

Definition:

Univariate plots are graphical methods used to analyze the distribution and characteristics of a **single variable**. They help in understanding central tendency, spread, skewness, and outliers before moving to bivariate or multivariate analysis.

Types of Univariate Plots

1. Histogram

- Displays frequency distribution of continuous data.
- Bars touch each other since data is continuous.
- Useful for detecting skewness and spread.

2. Box Plot (Whisker Plot)

- Summarizes data using **minimum, Q1, median, Q3, maximum**.
- Highlights **outliers** clearly.
- Useful for comparing distributions across categories.

3. **Density Plot**

- Smooth curve showing probability distribution of a variable.
- Provides clearer visualization of distribution shape than histogram.

4. **Bar Plot**

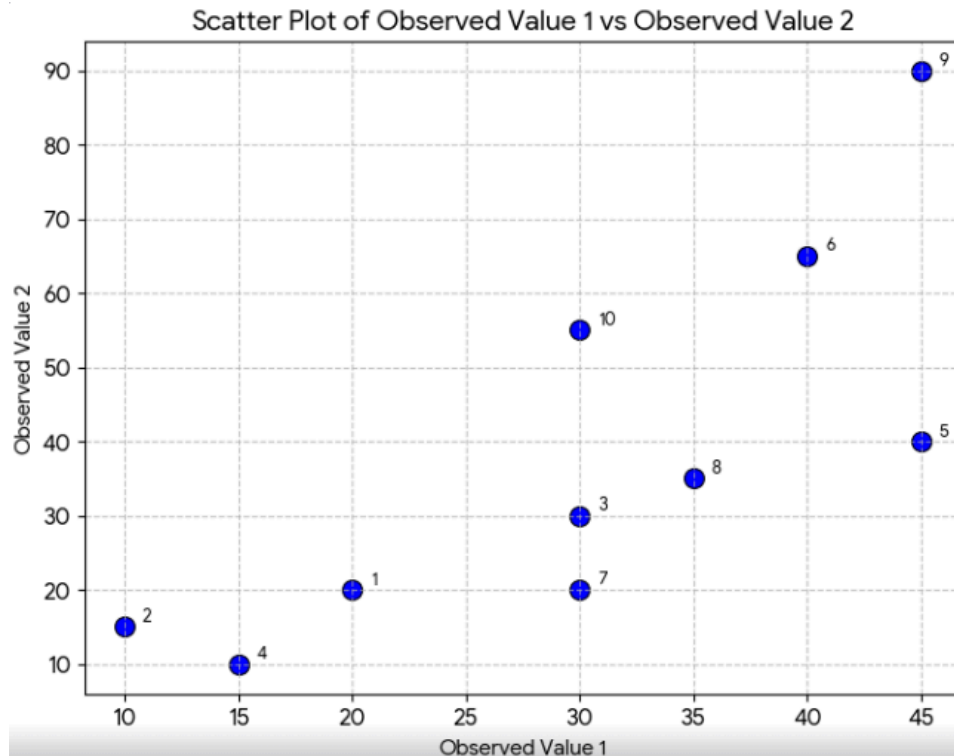
- Used for categorical variables.
- Shows frequency or proportion of each category.

5. **Pie Chart**

- Represents categorical data as proportions of a whole.
- Less preferred in EDA due to difficulty in comparing slices.

Q4. [10 Marks each] - Answers

a. What are the various challenges in WSN and explain the architecture of WSN?



Definition

A **scatter plot** is a graphical method used to study the relationship between two numerical variables. The **correlation coefficient** measures the strength and direction of that relationship.

Observation from Scatter Plot

- The points generally move **upward from left to right**.
- This indicates a **positive relationship** between Observed Value 1 and Observed Value 2.
- As Observed Value 1 increases, Observed Value 2 also tends to increase.

Coefficient of Correlation

The Pearson correlation coefficient is:

$$r=0.7766$$

Interpretation

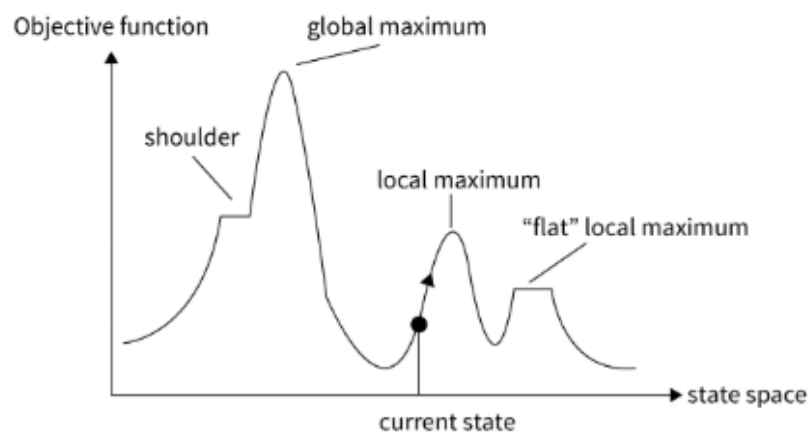
- Since $r \approx 0.78$, there is a **strong positive correlation** between the two variables.
- Correlation values range from:
 - $+1 \rightarrow$ perfect positive correlation
 - $0 \rightarrow$ no correlation
 - $-1 \rightarrow$ perfect negative correlation

Thus, the computed correlation confirms the observation from the scatter plot.

b. Explain the working of the Hill climbing algorithm. What are issues in Hill climbing?

Definition:

Hill Climbing is a heuristic search algorithm that continuously moves towards the neighbor state with the highest value (best solution) until no better neighbor exists. It is a **greedy local search** technique.



Working of Hill Climbing Algorithm

1. **Start State:** Begin with an initial solution (random or chosen).
2. **Evaluation Function:** Measure the quality (fitness/height) of the current state.
3. **Generate Neighbors:** Create small variations of the current state.
4. **Selection:** Choose the neighbor with the highest evaluation value.

5. **Iteration:** Repeat the process until no better neighbor exists.
6. **Termination:** Stops when a peak (local maximum) is reached or no better move is possible.

The algorithm is like climbing a hill: always moving upward until you cannot go higher.

Issues in Hill Climbing

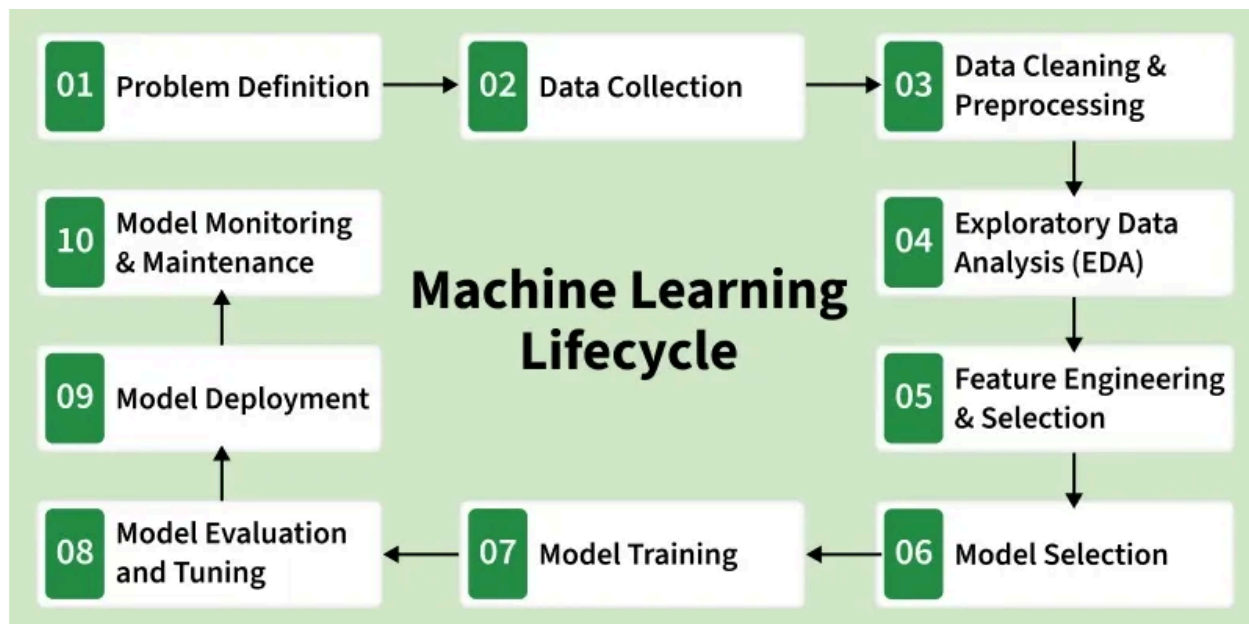
- **Local Maxima:** Algorithm may get stuck at a peak that is not the global optimum.
- **Plateaus:** Flat regions where all neighbors have the same value → search stalls.
- **Ridges:** Narrow paths requiring complex moves; simple hill climbing fails.
- **No Backtracking:** Once stuck, it cannot explore alternative paths.
- **Incomplete:** Does not guarantee finding the global optimum.
- **Sensitive to Initial State:** Different starting points may lead to different outcomes.

Q5. [10 Marks each] - Answers

a. Describe steps for developing ML applications with a labeled diagram.

Definition:

Developing ML applications involves a systematic process of transforming raw data into a trained model that can make predictions or decisions.



Main Steps

1. Problem Definition

- Clearly state the objective (classification, regression, clustering).
- Identify input variables and expected output.

2. Data Collection

- Gather relevant data from databases, sensors, APIs, or files.
- Ensure sufficient quantity and quality.

3. Data Preprocessing

- Clean data (handle missing values, remove duplicates).
- Transform features (normalization, encoding).
- Reduce dimensionality if needed.

4. Model Selection

- Choose suitable algorithm (e.g., Decision Tree, SVM, Neural Network).
- Consider problem type and data characteristics.

5. Model Training

- Feed training data into the algorithm.

- Adjust parameters to learn patterns.

6. Model Evaluation

- Test model on validation/test data.
- Use metrics like accuracy, precision, recall, RMSE.

7. Deployment

- Integrate model into production environment (apps, APIs).
- Ensure scalability and usability.

8. Monitoring & Maintenance

- Continuously track performance.
- Update/retrain model with new data.

b. What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.

Different Types of Environments

1. Fully Observable vs. Partially Observable

- Fully Observable: Agent has complete information (e.g., Chess).
- Partially Observable: Agent has limited information (e.g., Poker).

2. Deterministic vs. Stochastic

- Deterministic: Next state is predictable (e.g., solving a maze).
- Stochastic: Involves randomness (e.g., rolling dice).

3. Episodic vs. Sequential

- Episodic: Each task is independent (e.g., image classification).
- Sequential: Current decision affects future states (e.g., driving a car).

4. Static vs. Dynamic

- Static: Environment does not change while agent is thinking (e.g., crossword puzzle).

- Dynamic: Environment changes continuously (e.g., stock market).

5. Discrete vs. Continuous

- Discrete: Finite states/actions (e.g., Tic-Tac-Toe).
- Continuous: Infinite states/actions (e.g., robot arm movement).

6. Single Agent vs. Multi-Agent

- Single Agent: One agent acts alone (e.g., solving Sudoku).
- Multi-Agent: Multiple agents interact (e.g., football game).

Vacuum World Problem

- **Environment Description:**
 - A simple world with two rooms (A and B).
 - Each room may be clean or dirty.
 - Agent (vacuum cleaner) can move left, right, or suck dirt.
- **Environment Type:**
 - **Partially Observable:** Agent senses only current room.
 - **Deterministic:** Actions have predictable outcomes.
 - **Sequential:** Cleaning decisions affect future states.
 - **Dynamic:** Dirt may reappear.
 - **Discrete:** Finite states (clean/dirty, left/right).
 - **Single Agent:** Only one vacuum agent.
- **Example State Space:**
 - Initial State: Room A dirty, Room B clean, agent in A.
 - Goal State: Both rooms clean.
 - Operators: Move left/right, suck dirt.

Q6. [5 Marks each] - Answers

a. Write Short Note on: ANOVA

Definition:

ANOVA is a **statistical technique** used to compare the **means of three or more groups** to determine if there are significant differences among them.

Steps in ANOVA

1. State Hypotheses:

- Null hypothesis (H₀): All group means are equal.
- Alternative hypothesis (H₁): At least one group mean differs.

2. Calculate Group Means and Overall Mean.

3. Compute Sum of Squares:

- **Between-group variance (SSB)** — variation due to differences between group means.
- **Within-group variance (SSW)** — variation within each group.

4. Calculate F-Ratio:

$$F = \frac{\text{Mean Square Between (MSB)}}{\text{Mean Square Within (MSW)}}$$

5. Compare F-value with Critical Value:

- If $F > F_{\text{critical}}$, reject H₀.

6. Draw Conclusion:

- Conclude whether group means differ significantly.

b. Write Short Note on: Min Max Algorithm

Definition:

The **Min-Max Algorithm** is a decision-making algorithm used in **game theory and**

AI, particularly for **two-player, zero-sum games** (like Chess, Tic-Tac-Toe). It minimizes the possible loss for a worst-case scenario while maximizing the possible gain.

Working Principle

- **Players:**
 - **MAX player** → Tries to maximize score.
 - **MIN player** → Tries to minimize score.
- The algorithm explores the **game tree**:
 - At MAX nodes → choose the move with the highest value.
 - At MIN nodes → choose the move with the lowest value.
- Recursively evaluates all possible moves until terminal states are reached.

Steps

1. Generate the game tree up to a certain depth.
2. Apply evaluation function at leaf nodes.
3. Propagate values upward:
 - MAX node → select maximum child value.
 - MIN node → select minimum child value.
4. Root node decision → best move for MAX player.

Example

- In Tic-Tac-Toe:
 - MAX (X) tries to win.
 - MIN (O) tries to block or minimize X's chances.
 - Min-Max evaluates all possible moves and selects the optimal one.

c. Write Short Note on: Non-graphical EDA

Definition:

Non-graphical EDA refers to techniques that summarize and analyze data **without using visual plots or charts**. Instead, it relies on **numerical and tabular methods** to describe data characteristics.

Key Techniques in Non-Graphical EDA**1. Summary Statistics**

- Measures of central tendency: Mean, Median, Mode.
- Measures of dispersion: Range, Variance, Standard Deviation.

2. Frequency Tables

- Shows how often each value or category occurs.
- Useful for categorical data analysis.

3. Cross-tabulation

- Tabular representation of relationships between two variables.
- Example: Gender vs Purchase Decision.

4. Numerical Measures of Shape

- Skewness (asymmetry of distribution).
- Kurtosis (peakedness of distribution).

5. Correlation & Covariance

- Quantifies relationships between variables numerically.

d. Write Short Note on: Forward chaining based proofs**Definition:**

Forward chaining is a **data-driven inference technique** used in rule-based systems. It starts from known facts and applies inference rules to derive new facts until the goal (proof) is reached.

Working of Forward Chaining Proofs

- 1. Start with Known Facts:** Begin with the initial knowledge base.

2. **Apply Rules:** Use inference rules of the form *IF condition THEN conclusion*.
3. **Generate New Facts:** Add conclusions to the knowledge base.
4. **Repeat:** Continue applying rules until the desired goal or proof is obtained.

Role in Proofs

- Used to **prove theorems** by expanding facts step by step.
- Suitable for **prediction and simulation** problems.
- Common in **expert systems** (e.g., medical diagnosis).

Example

- Rule 1: IF Human(x) $\rightarrow$ Mortal(x)
- Fact: Human(Socrates)
- Forward Chaining Proof:
 - Apply Rule 1 $\rightarrow$ Mortal(Socrates).
 - Goal achieved: Socrates is mortal.

e. Write Short Note on: Box Plot

Definition:

A **Box Plot** (or Whisker Plot) is a graphical representation of the **distribution of a dataset** based on five summary statistics: minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.

Key Features

- **Box:** Represents the **interquartile range (IQR = Q3 – Q1)**.
- **Median Line:** Inside the box, shows the central tendency.
- **Whiskers:** Extend to minimum and maximum values (excluding outliers).
- **Outliers:** Plotted as individual points beyond whiskers.

Role in EDA

- Detects **spread, skewness, and variability** of data.

- Identifies **outliers** quickly.
- Useful for comparing distributions across multiple groups.

Example

- Exam scores of students:
 - Box shows middle 50% of scores.
 - Median line shows typical score.
 - Outliers highlight unusually high/low performers.

AIDS 2025 December PYQ Answers

Q1. [5 Marks each]

- a)** Differentiate between Simple Reflex Agent and Model-Based Reflex Agent.
- b)** Discuss Problem Formulation in AI with an example.
- c)** Define heuristic search. Give two examples of heuristic functions.
- d)** Describe the process of knowledge representation using First Order Logic.

Q2. [10 Marks each]

- a)** What are Supervised Machine Learning techniques? Explain Logistic Regression technique. What makes it a Supervised learning technique?
- b)** Compare and contrast univariate and multivariate EDA with suitable examples.

Q3. [10 Marks each]

- a)** Explain the basic steps in developing a Machine Learning application.
- b)** Explain the importance of graphical methods in EDA.

Q4. [10 Marks each]

- a)** Describe in detail the stages in the Data Science Lifecycle.
- b)** Discuss Partial Order, Hierarchical, and Conditional Planning.

Q5. [10 Marks each]

- a)** Compare and contrast Data Science, Business Analytics, and Big Data with examples.
- b)** Define Unsupervised Machine Learning techniques. Discuss K-Means and Hierarchical Clustering in detail.

Q6. [10 Marks each]

a) What do you mean by covariance and correlation? Explain their interpretation and the range of coefficients.

Experiment No.	1	2	3	4	5	6	7	8	9	10
X (Variable 1)	10	12	15	18	20	22	25	30	35	40
Y (Variable 2)	8	11	14	17	21	24	28	32	36	39

For the above data, calculate Covariance (COV) and Correlation Coefficient (CORR) between X and Y. Interpret your results.

b) Explain the working of the Hill Climbing algorithm. What are issues in Hill Climbing?

Q1. [5 Marks each] - Answers

a. Differentiate between Simple Reflex Agent and Model-Based Reflex Agent.

Basis	Simple Reflex Agent	Model-Based Reflex Agent
Decision Basis	Uses only current percept	Uses current percept + past information
Internal State	No internal memory/state	Maintains internal state (model)
Environment Handling	Works well in fully observable environments	Works in partially observable environments
Complexity	Simple and fast	More complex
Adaptability	Less flexible	More flexible
Example	Thermostat switching on/off	Robot tracking objects using memory

b. Discuss Problem Formulation in AI with an example.

Definition:

Problem formulation in AI is the process of defining a problem in a structured way so that it can be solved by a search algorithm.

Key Points:

- **Initial State:** Starting point of the agent.
 - **State Space:** All possible states reachable from the initial state.
 - **Actions (Operators):** Possible moves from one state to another.
 - **Transition Model:** Defines result of applying an action to a state.
 - **Goal State:** Desired final state the agent wants to reach.
 - **Path Cost:** Cost associated with reaching the goal (e.g., distance, time).
 - Converts real-world problem into a **search problem**.
-

Example (8-Puzzle Problem):

- **Initial State:** Random arrangement of tiles
- **State Space:** All possible tile arrangements
- **Actions:** Move blank tile up/down/left/right
- **Goal State:** Tiles arranged in order
- **Path Cost:** Number of moves taken to solve

c. Define heuristic search. Give two examples of heuristic functions.

Definition:

Heuristic search is a search technique that uses a **heuristic function ($h(n)$)** to estimate the best path to reach the goal, making the search faster and more efficient.

Key Points:

- Uses **problem-specific knowledge** to guide the search.
- Heuristic function estimates **cost from current state to goal**.

- Reduces **search space** compared to uninformed search.
- Does not always guarantee optimal solution (depends on heuristic).
- Commonly used in algorithms like **A*** and Greedy Best-First Search.
- Improves efficiency by exploring **more promising paths first**.

Examples of Heuristic Functions:

1. Manhattan Distance:

- Used in grid problems (like 8-puzzle).
- Sum of horizontal + vertical distances of tiles from goal position.

2. Euclidean Distance:

- Straight-line distance between current state and goal.
- Commonly used in pathfinding problems (like maps).

d. Describe the process of knowledge representation using First Order Logic.

Definition:

First-Order Logic (FOL) is a formal system used in AI to represent knowledge with objects, relations, and quantifiers, enabling reasoning about the world.

Process Steps:

1. **Identify Objects:** Define entities in the domain (e.g., humans, animals, cars).
2. **Define Predicates:** Represent properties or relationships (e.g., Human(x), Loves(x,y)).
3. **Use Quantifiers:**
 - Universal ($\forall$): Statement applies to all objects.
 - Existential ($\exists$): Statement applies to at least one object.
4. **Form Sentences:** Combine predicates, quantifiers, and logical connectives ($\wedge, \vee, \rightarrow, \neg$).

5. **Knowledge Base:** Store these logical sentences for reasoning.

6. **Inference:** Apply rules (like resolution) to derive new facts or check queries.

Example:

- Knowledge: "All humans are mortal. Socrates is a human."
- Representation:
 - $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$
 - $\text{Human}(\text{Socrates})$
- Query: Is Socrates mortal?
- Inference: From the rules, we conclude $\text{Mortal}(\text{Socrates})$.

Q2. [10 Marks each] - Answers

a. What are Supervised Machine Learning techniques? Explain Logistic Regression technique. What makes it a Supervised learning technique?

Definition:

Supervised Machine Learning techniques are methods where a model is trained on **labeled data** (input + correct output) to learn a mapping and make predictions.

Key Points:

- Includes **classification and regression** techniques.
- Learns from **input-output pairs**.
- Common techniques: Linear Regression, Logistic Regression, Decision Trees, SVM, KNN.
- Goal is to **predict output for new unseen data**.

Logistic Regression:

- Used for **classification problems** (mainly binary: 0 or 1).

- Uses a **sigmoid function** to convert output into probability (0 to 1).
- If probability $\geq 0.5 \rightarrow$ class 1, else class 0.
- Models relationship between input features and **probability of class**.
- Despite name, it is used for **classification, not regression**.

What makes it Supervised:

- Trained using **labeled data** (input + known class).
- Learns from **actual outcomes** during training.
- Adjusts parameters to **minimize prediction error**.

Example:

Predicting whether an email is spam (1) or not spam (0) based on labeled training data.

b. Compare and contrast univariate and multivariate EDA with suitable examples.

Aspect	Univariate EDA	Multivariate EDA
Definition	Analysis of a single variable	Analysis of two or more variables together
Purpose	Understand distribution, central tendency, spread	Understand relationships, correlations, interactions
Number of Variables	One	Two or more
Focus	Individual variable behavior	Relationship between variables
Techniques Used	Histogram, Box Plot, Bar Chart	Scatter Plot, Heatmap, Pair Plot
Complexity	Simple	More complex
Insights	Mean, median, mode, variance, outliers	Correlation, trends, patterns, dependencies
Type of Analysis	Descriptive	Comparative and relational

Aspect	Univariate EDA	Multivariate EDA
Example	Distribution of student marks in one subject	Relationship between study hours and marks

Q3. [10 Marks each] - Answers

b. Explain the importance of graphical methods in EDA.

Definition:

Graphical methods in Exploratory Data Analysis (EDA) are visual techniques used to summarize and understand data distributions, patterns, and relationships. They provide intuitive insights that numerical summaries alone cannot reveal.

Main Explanation

1. Data Distribution Visualization

- Graphs like histograms and density plots show how values are spread (normal, skewed, uniform).

2. Detection of Outliers

- Box plots highlight extreme values that may affect analysis.

3. Understanding Relationships

- Scatter plots and pair plots reveal correlations between variables.

4. Categorical Analysis

- Bar plots and pie charts help compare frequencies across categories.

5. Trend Identification

- Line charts show changes over time (time-series data).

6. Simplifies Complex Data

- Converts large datasets into easy-to-understand visuals.

7. Supports Decision Making

- Clear visuals help stakeholders interpret results quickly.

Q4. [10 Marks each] - Answers

a. Describe in detail the stages in the Data Science Lifecycle.

Definition:

The Data Science Lifecycle is a structured process that guides the development of data-driven solutions, from problem identification to deployment and monitoring.

Main Stages

1. Problem Definition

- Identify business or research objectives.
- Define input variables and expected outcomes.

2. Data Acquisition

- Collect data from databases, APIs, sensors, or external sources.
- Ensure relevance and adequacy of data.

3. Data Preprocessing

- Clean data (handle missing values, duplicates).
- Integrate data from multiple sources.
- Transform features (normalization, encoding).

4. Exploratory Data Analysis (EDA)

- Use statistical and graphical methods to understand distributions, trends, and outliers.
- Identify correlations and patterns.

5. Feature Engineering & Selection

- Create new features or select important ones.
- Reduce dimensionality to improve efficiency.

6. Model Building

- Choose appropriate algorithms (classification, regression, clustering).

- Train models using training datasets.

7. Model Evaluation

- Test model on validation/test data.
- Use metrics like accuracy, precision, recall, RMSE, F1-score.

8. Deployment

- Integrate the model into applications, APIs, or dashboards.
- Ensure scalability and usability.

9. Monitoring & Maintenance

- Track performance in real-world use.
- Retrain/update model with new data to maintain accuracy.

b. Discuss Partial Order, Hierarchical, and Conditional Planning.

Definition:

Planning in AI is the process of generating a sequence of actions that an agent must perform to achieve a goal. Different techniques handle complexity, uncertainty, and flexibility in planning.

1. Partial Order Planning

- **Concept:** Plans actions without fixing their exact order initially.
- **Key Idea:** Only specify ordering constraints where necessary.
- **Advantage:** Allows parallel execution of independent actions.
- **Example:** Cooking → chopping vegetables and boiling water can be done in parallel before combining.

2. Hierarchical Planning

- **Concept:** Breaks down complex goals into smaller sub-goals.
- **Key Idea:** Uses abstraction levels (high-level → low-level tasks).
- **Advantage:** Simplifies planning by handling tasks step by step.

- **Example:** Planning a trip → Book tickets → Arrange accommodation → Plan sightseeing.

3. Conditional Planning

- **Concept:** Considers uncertainty and alternative paths depending on conditions.
- **Key Idea:** Plan includes “if-then” branches for different scenarios.
- **Advantage:** Handles unpredictable environments.
- **Example:** Delivery agent → “If traffic jam → take alternate route; else → follow main road.”

Q5. [10 Marks each] - Answers

b. Define Unsupervised Machine Learning techniques. Discuss K-Means and Hierarchical Clustering in detail.

Definition:

Unsupervised Machine Learning techniques are methods where models learn patterns from **unlabeled data**. The aim is to discover hidden structures, clusters, or relationships without predefined categories.

K-Means Clustering

- **Concept:** Partitioning algorithm that divides data into k clusters based on nearest centroid.
- **Steps:**
 1. Choose number of clusters k.
 2. Initialize centroids randomly.
 3. Assign each data point to the nearest centroid.
 4. Update centroids as mean of assigned points.
 5. Repeat until centroids stabilize.
- **Advantages:** Simple, fast, scalable.

- **Limitations:** Sensitive to initial centroid choice and outliers.
- **Example:** Segmenting customers into 3 groups based on income and spending score.

Hierarchical Clustering

- **Concept:** Builds a hierarchy of clusters using **Agglomerative (bottom-up)** or **Divisive (top-down)** approaches.
- **Steps:**
 1. Start with each point as a separate cluster.
 2. Merge closest clusters (Agglomerative) OR split clusters (Divisive).
 3. Continue until one cluster or desired number is reached.
- **Distance Measures:** Single linkage, complete linkage, average linkage.
- **Advantages:** No need to predefine k, provides dendrogram for visualization.
- **Limitations:** Computationally expensive for large datasets.
- **Example:** Grouping species based on genetic similarity using dendrogram.

Q6. [10 Marks each] - Answers

a. What do you mean by covariance and correlation? Explain their interpretation and the range of coefficients. For the above data, calculate Covariance (COV) and Correlation Coefficient (CORR) between X and Y. Interpret your results.

Covariance

- Measures how two variables vary together.
- $$Cov(X, Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n-1}$$
- **Interpretation:**
 - Positive → both increase/decrease together.

- Negative → one increases while the other decreases.
- Zero → no linear relationship.
- **Range:** Any real value ($-\infty$ to $+\infty$).

Correlation

- Standardized measure of strength and direction of linear relationship.
- $r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$
- **Interpretation:**
 - +1 → perfect positive correlation.
 - -1 → perfect negative correlation.
 - 0 → no correlation.
- **Range:** Always between -1 and +1.

Covariance Calculation

Calculated covariance:

$$Cov(X,Y)=103.89$$

Correlation Coefficient Calculation

Calculated correlation coefficient:

$$r=0.989$$

Interpretation of Results

Covariance

- Covariance is **positive (103.89)**.
- This means X and Y move in the **same direction**.
- As X increases, Y also increases.

Correlation

- $r=0.989$ is very close to +1.

- This indicates a **very strong positive linear relationship** between X and Y.

AIDS 2024 May PYQ Answers

Q1. [5 Marks each]

- a) Formulate the problem for 8 queens.
- b) What is the difference between unification and Skolemization?
- c) Define over fitting and under fitting in detail with a diagram.
- d) Explain utility based agent architecture with diagram.
- e) Compare box-plot and scatter-plot.
- f) Explain working of Forward and Backward Chaining algorithms.

Q2. [10 Marks each]

- a) Consider following 5×5 grid where S represents start and G goal position, # represents an obstacle, .(dot) represents free to move in cell. Consider agent can follow four standard moves in this puzzle world. Represent this puzzle as state space search problem. Apply Hill Climbing Search. Does the algorithm stuck in Local Minima?

S	.	.	#	.
#	.	#	.	.
.	.	.	#	.
.	#	.	.	.
.	.	.	.	G

- b) Write note on following supervised learning techniques:

- i) SVM
- ii) ID3

Q3. [10 Marks each]

- a) What do you mean by data analytics? What are the different types of data analytics?
- b) What is linear regression? Explain its significance in ML. Compare it with logistic regression.

Q4. [10 Marks each]

a) Perform t-test on following data about choice of customers who preferred either tea or coffee. The experiment was repeated 5 times and results are tabulated below. Comment whether the mean of two sets are same based on test result? (Consider $\alpha=0.05$. $t(0.05,3)=3.182$; $t(0.05,4)=2.776$; $t(0.05,8)=2.306$; $t(0.05,9)=2.262$; $t(0.05,10)=2.228$). Clearly mention Null and Alternate hypothesis. Also comment whether t-test fails or succeeds in rejecting Null Hypothesis.

Tea	4	5	7	6	9
Coffee	3	8	6	14	7

b) What are the different univariate plots in EDA? Explain them in detail.

Q5. [10 Marks each]

a) What are the different issues in ML algorithms?

b) Compare Z-Test, T-Test and ANOVA in detail.

Q6. [10 Marks each]

a) Describe the architecture of ML application. Explain with a diagram.

b) Describe any four uninformed search strategies. Compare them with time complexity, space complexity, optimality and completeness.

Q1. [5 Marks each] - Answers

a. Formulate the problem for 8 queens.

Definition:

The **8 Queens Problem** is to place 8 queens on a chessboard such that no two queens attack each other.

Formulation Components:

- **Initial State:** An empty 8×8 chessboard.
- **State Space:** All possible arrangements of queens on the board.
- **Operators (Actions):** Place a queen in a valid position (row/column).
- **Goal State:** 8 queens placed so that none share the same row, column, or diagonal.
- **Path Cost:** Not relevant (any valid solution is acceptable).

b. What is the difference between unification and Skolemization?

Basis	Unification	Skolemization
Definition	Process of making two logical expressions identical by finding substitutions	Process of removing existential quantifiers by introducing Skolem constants/functions
Purpose	Used in inference to match predicates	Used in converting FOL into standard (clausal) form
Operation	Finds substitutions for variables	Replaces variables with constants/functions
Quantifiers	Does not remove quantifiers	Eliminates existential ($\exists$) quantifiers
Use Case	Used in resolution and theorem proving	Used during preprocessing before resolution
Output	Substitution set (e.g., $\{x = \text{John}\}$)	Modified expression without $\exists$
Example	Unify $P(x)$ and $P(\text{John}) \rightarrow \{x = \text{John}\}$	$\exists x P(x) \rightarrow P(a)$ (a is Skolem constant)

c. Define over fitting and under fitting in detail with a diagram.

Definition:

- **Overfitting:** Model learns training data too closely, including noise, leading to poor generalization.
- **Underfitting:** Model is too simple, fails to capture patterns, resulting in poor accuracy.

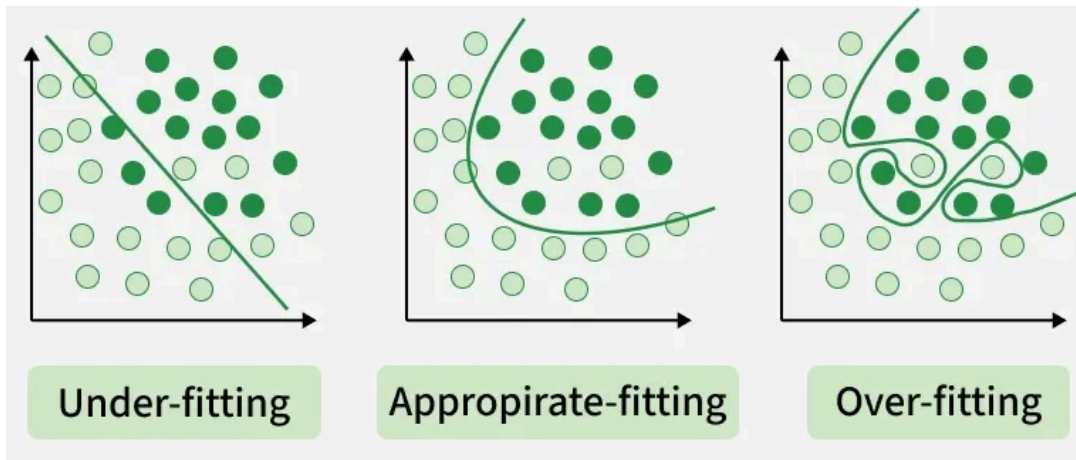
Effects of Overfitting:

- High accuracy on training data but low accuracy on test data.
- Model memorizes noise and irrelevant details.
- Poor generalization to unseen datasets.
- Misleadingly optimistic training results.
- Needs regularization, pruning, or more data to fix.

Effects of Underfitting:

- Low accuracy on both training and test data.
- Model ignores important relationships in data.
- Predictions are overly simplistic.
- High bias, low variance.

- Improved by using complex models or better features.



d. Explain utility based agent architecture with diagram

Definition:

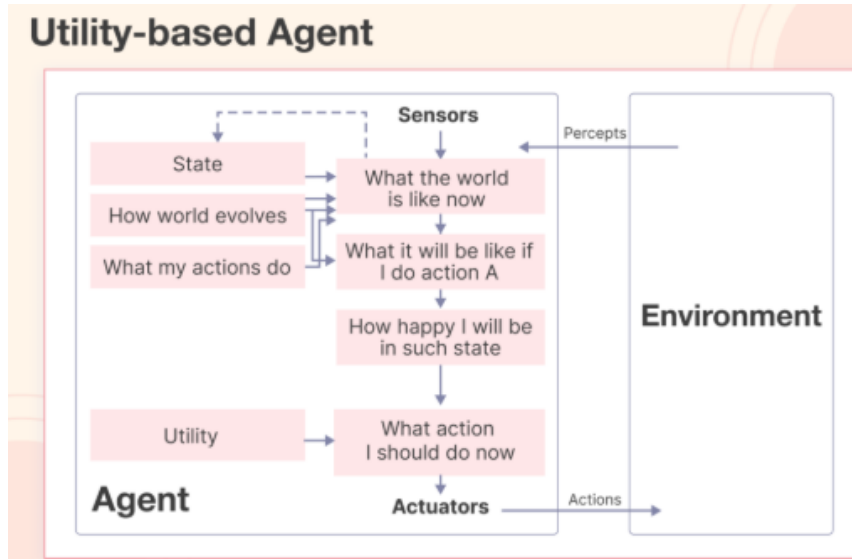
A **Utility-Based Agent** is an intelligent agent that chooses actions based on a utility function, which measures how desirable a particular state is, aiming to maximize overall performance.

Key Points:

- **Utility Function:** Assigns numerical values to states, representing preference or desirability.
- **Decision Making:** Selects actions that lead to states with higher utility.
- **Flexibility:** Can handle conflicting goals by comparing utilities.
- **Adaptability:** Works well in complex, uncertain environments.
- **Improvement over Goal-Based Agents:** Not only achieves goals but also optimizes outcomes.
- **Applications:** Autonomous vehicles, recommendation systems, resource allocation.

Example:

- In **autonomous driving**, the agent may have multiple possible routes. A utility function evaluates each route based on safety, time, and fuel efficiency, and the agent selects the route with the highest utility.



e. Compare box-plot and scatter-plot.

Basis	Box-Plot	Scatter-Plot
Definition	Shows distribution of a single variable using quartiles	Shows relationship between two variables
Data Type	Univariate (one variable)	Bivariate (two variables)
Purpose	Identify spread, median, and outliers	Identify correlation or pattern between variables
Representation	Box with whiskers (min, Q1, median, Q3, max)	Points plotted on X-Y plane
Outliers	Clearly visible as separate points	Can be seen but not explicitly highlighted
Use Case	Compare distributions across groups	Analyze trends or relationships
Example	Distribution of student marks	Relation between height and weight

f. Explain working of Forward and Backward Chaining algorithms.

Definition:

Both are reasoning techniques in AI used with rule-based systems to derive conclusions.

Forward Chaining (Data → Goal):

- Starts with known facts.
- Applies inference rules to generate new facts.

- Continues until the goal is reached.
- Suitable for **prediction and simulation**.
- Example: From symptoms → predict disease.

Backward Chaining (Goal → Data):

- Starts with a goal (hypothesis).
- Works backward to check if supporting facts exist.
- Uses rules to verify conditions.
- Suitable for **diagnosis and problem-solving**.
- Example: From suspected disease → verify symptoms.

Q2. [10 Marks each] - Answers

a. Consider following 5x5 grid where S represents start and G goal position, # represents an obstacle, .(dot) represents free to move in cell. Consider agent can follow four standard moves in this puzzle world. Represent this puzzle as state space search problem. Apply Hill Climbing Search. Does the algorithm stuck in Local Minima?

Introduction

A **state space search problem** represents a problem using:

- Initial state
- Goal state
- Valid actions/operators
- State transitions
- Goal test

Hill Climbing is a local search algorithm that always moves toward the state that appears closer to the goal.

1. State Space Representation

Components of State Space

1. States

- Each cell in the 5×5 grid represents a state.

- A state is represented as:

```
(row, column)
```

Example:

```
Start State = (1,1)  
Goal State = (5,4)
```

2. Initial State

```
Initial State = S
```

The agent starts from cell **S**.

3. Goal State

```
Goal State = G
```

The agent must reach cell **G**.

4. Operators / Actions

The agent can move in four directions:

- Up
- Down
- Left
- Right

Rules:

- Movement allowed only in free cells ()
- Obstacle cells () cannot be entered

5. State Transition

Moving from one valid cell to another valid neighboring cell.

Example:

$(1,1) \rightarrow (1,2)$

6. Goal Test

Check whether current state equals goal state.

Current State = Goal State

2. Applying Hill Climbing Search

Heuristic Function

Hill Climbing uses a heuristic function:

$h(n) = \text{Distance from current state to goal}$

Using Manhattan Distance:

$h(n) = |x_g - x_n| + |y_g - y_n|$

Where:

- (x_g, y_g) = Goal position
- (x_n, y_n) = Current position

Working of Hill Climbing

Step 1

Start from:

$S = (1,1)$

Step 2

Evaluate neighboring cells and choose the one with minimum heuristic value.

Possible moves:

- Right
- Down

Obstacle cells are ignored.

Step 3

Continue moving to the neighboring state that appears closer to Goal.

The algorithm greedily moves toward **G**.

3. Does Hill Climbing Get Stuck in Local Minima?

Yes, Hill Climbing can get stuck in Local Minima.

Reason

- Obstacles block the direct path toward Goal.
 - Hill Climbing always selects the immediately better state.
 - Sometimes reaching the Goal requires moving temporarily away from the Goal.
 - Hill Climbing does not perform backtracking.
-

In This Grid

The agent may reach a state where:

- All neighboring states have worse or equal heuristic values.
- Obstacles prevent further progress.

Thus the algorithm stops before reaching Goal.

This situation is called:

Local Minima

A state that is better than all neighboring states but is not the actual goal state.

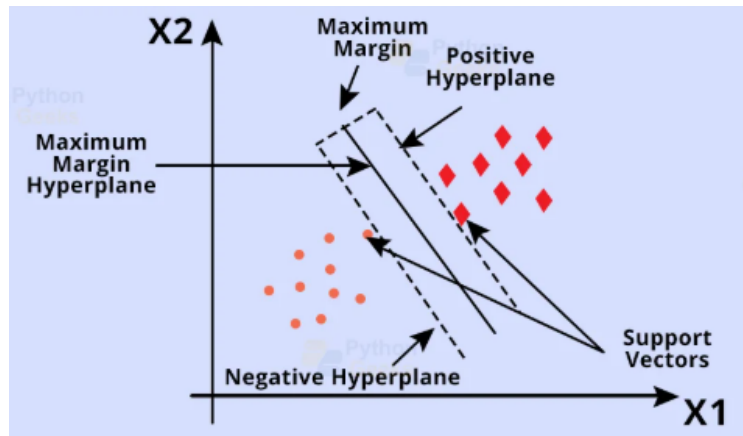
b. Write note on following supervised learning techniques: i) SVM & ii) ID3

i) Support Vector Machine (SVM)

SVM is a supervised learning algorithm used mainly for **classification**. It finds the **optimal hyperplane** that separates data points of different classes with maximum margin.

Key Concepts

- **Hyperplane**: Decision boundary separating classes.
- **Margin**: Distance between hyperplane and nearest data points.
- **Support Vectors**: Data points closest to the hyperplane.
- **Kernel Trick**: Transforms data into higher dimensions (linear, polynomial, RBF).



Working

1. Plot data points in feature space.
2. Find a hyperplane that separates classes.
3. Maximize margin between classes.
4. Use kernel if data is not linearly separable.

Example

- Email classification: Spam vs Not Spam.

ii) ID3 (Iterative Dichotomiser 3)

ID3 is a **decision tree algorithm** used for classification. It builds a tree by selecting attributes based on **Information Gain**.

Key Concepts

- **Entropy**: Measures impurity in data.
- **Information Gain**: Reduction in entropy after splitting.
- **Root Node**: Attribute with highest information gain.

Working

1. Calculate entropy of dataset.
2. Compute information gain for each attribute.
3. Select attribute with highest gain as root.
4. Split dataset based on that attribute.
5. Repeat recursively until all data is classified.

Example

- Predict whether a person will play tennis based on weather conditions.

Q3. [10 Marks each] - Answers

a. What do you mean by data analytics? What are the different types of data analytics?

Definition:

Data Analytics is the process of examining raw data to extract meaningful insights, identify patterns, and support decision-making. It transforms data into actionable knowledge for businesses, science, and technology.

Types of Data Analytics

1. Descriptive Analytics

- Explains *what happened* using historical data.
- Example: Monthly sales reports showing revenue trends.

2. Diagnostic Analytics

- Explains *why it happened* by finding causes.
- Example: Analyzing drop in sales due to reduced customer visits.

3. Predictive Analytics

- Forecasts *what is likely to happen* using statistical models and machine learning.
- Example: Predicting customer churn based on past behavior.

4. Prescriptive Analytics

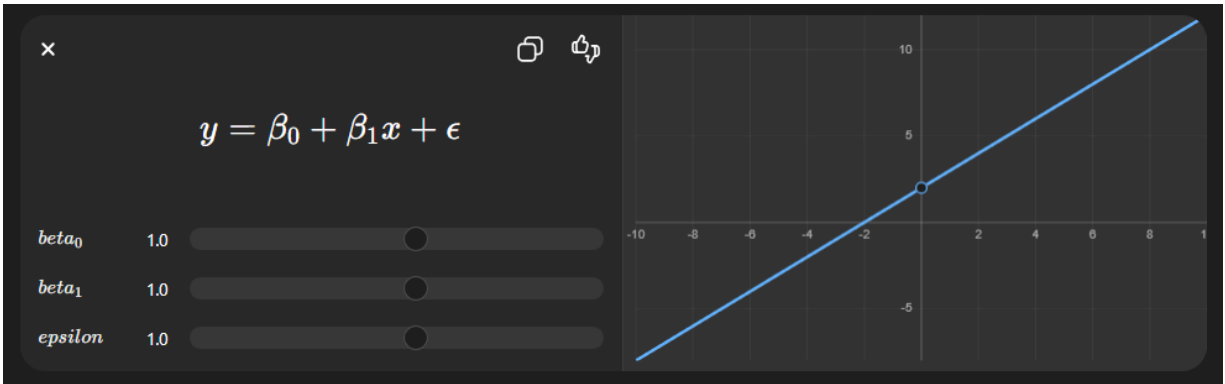
- Suggests *what should be done* for best outcomes.
- Example: Recommending discounts or promotions to retain customers.

b. What is linear regression? Explain its significance in ML. Compare it with logistic regression.

Definition

Linear Regression is a supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a straight line.

Linear Regression



Explanation

- Predicts **continuous values** (e.g., price, marks).
- Finds the best-fit line by minimizing error (usually using least squares).
- Coefficients (β_0, β_1) represent intercept and slope.

$\beta_0, \beta_1 \backslash \text{beta}_0, \text{beta}_1$

Significance in ML

- Simple and easy to interpret.
- Acts as a **baseline model** for regression problems.
- Helps understand **feature relationships**.
- Computationally efficient.
- Widely used in forecasting and trend analysis.

Aspect	Linear Regression	Logistic Regression
Type of Problem	Regression	Classification
Output	Continuous values	Probabilities (0 to 1)
Model Equation	Linear equation	Sigmoid function
Output Range	$(-\infty)$ to $(+\infty)$	0 to 1
Goal	Predict numeric value	Predict class label
Error Function	Mean Squared Error	Log Loss (Cross-Entropy)
Assumption	Linear relationship	Linear decision boundary
Example	Predict house price	Predict spam or not spam

Q4. [10 Marks each] - Answers

a. Perform t-test on following data about choice of customers who preferred either tea or coffee. The experiment was repeated 5 times and results are tabulated below. Comment whether the mean of two sets are same based on test result? (Consider $\alpha=0.05$. $t(0.05,3)=3.182$; $t(0.05,4)=2.776$; $t(0.05,8)=2.306$; $t(0.05,9)=2.262$; $t(0.05,10)=2.228$). Clearly mention Null and Alternate hypothesis. Also comment whether t-test fails or succeeds in rejecting Null Hypothesis.

Definition

A **t-test** is a statistical hypothesis test used to compare the means of two datasets and determine whether the difference between them is statistically significant.

Given Data:

Tea	4	5	7	6	9
Coffee	3	8	6	14	7

Number of observations:

$$n_1 = n_2 = 5$$

Significance level:

$$\alpha = 0.05$$

Critical value given:

$$t(0.05,8) = 2.306$$

Step 1: State Hypotheses

Null Hypothesis (H_0)

The mean preference for Tea and Coffee is same.
 $\mu_1 = \mu_2$

Alternate Hypothesis (H_1)

The mean preference for Tea and Coffee is different.
 $\mu_1 \neq \mu_2$

Step 2: Calculate Means

Mean of Tea

$$\begin{aligned}\bar{X}_1 &= (4 + 5 + 7 + 6 + 9) / 5 \\ &= 31 / 5 \\ &= 6.2\end{aligned}$$

Mean of Coffee

$$\begin{aligned}\bar{X}_2 &= (3 + 8 + 6 + 14 + 7) / 5 \\ &= 38 / 5 \\ &= 7.6\end{aligned}$$

Step 3: Calculate Standard Deviations

For Tea

$$\begin{aligned}s_1^2 &= 3.7 \\ s_1 &= 1.92\end{aligned}$$

For Coffee

$$\begin{aligned}s_2^2 &= 16.8 \\ s_2 &= 4.10\end{aligned}$$

Step 4: Calculate t-value

Using independent sample t-test formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Substituting values:

$$\begin{aligned}t &= (6.2 - 7.6) / \sqrt{((3.7/5) + (16.8/5))} \\ t &= (-1.4) / \sqrt{(0.74 + 3.36)} \\ t &= (-1.4) / \sqrt{4.10}\end{aligned}$$

$$t = (-1.4) / 2.02$$

$$t \approx -0.693$$

Step 5: Compare with Critical Value

Calculated t-value:

$$|t| = 0.693$$

Critical value:

$$t_{\text{critical}} = 2.306$$

Comparison:

$$0.693 < 2.306$$

Step 6: Decision

Since calculated t-value is less than critical value:

Fail to reject Null Hypothesis (H_0)

Interpretation

- There is no significant difference between the mean preferences for Tea and Coffee.
- The observed difference may be due to random variation.

Conclusion

The t-test shows that the means of Tea and Coffee preferences are statistically the same at significance level $\alpha = 0.05$.

Therefore:

- Null Hypothesis is accepted.
- t-test fails to reject the Null Hypothesis.

Q5. [10 Marks each] - Answers

b. Compare Z-Test, T-Test and ANOVA in detail.

Aspect	Z-Test	T-Test	ANOVA (Analysis of Variance)
Definition	Tests difference between means using normal distribution	Tests difference between means using t-distribution	Tests difference among 3 or more group means
Number of Samples	One or two samples	One or two samples	Three or more samples
Population Variance	Known	Unknown	Unknown
Sample Size	Large ($n \geq 30$)	Small ($n < 30$)	Any (usually ≥ 3 groups)
Distribution Used	Normal (Z distribution)	t-distribution	F-distribution
Purpose	Compare sample mean with population mean or between two means	Compare means of small samples	Compare multiple group means simultaneously
Test Statistic	Z-value	t-value	F-value
Complexity	Simple	Moderate	More complex
Example	Test if average height equals 170 cm	Compare marks of two classes	Compare marks of 3 different classes
Assumptions	Normality, known variance	Normality, unknown variance	Normality, equal variances, independence

Q6. [10 Marks each] - Answers

a. Describe the architecture of ML application

Definition:

The architecture of an ML application refers to the structured design that integrates **data pipelines, model development, deployment, and monitoring** to deliver intelligent solutions.

Stages in ML Application Architecture

1. Data Layer

- **Sources:** Databases, APIs, sensors, files.
- **Functions:** Data collection, storage, preprocessing (cleaning, normalization).

2. Feature Engineering Layer

- Transform raw data into meaningful features.
- Includes dimensionality reduction, encoding, scaling.

3. Model Layer

- **Model Selection:** Choose algorithm (Regression, SVM, Neural Networks).
- **Training:** Fit model on training data.
- **Evaluation:** Validate using metrics (accuracy, precision, recall, RMSE).

4. Application Layer

- Integrates ML model into business applications (apps, dashboards, APIs).
- Provides user interface for interaction.

5. Deployment Layer

- Model hosted on cloud/on-premise servers.
- Accessible via REST APIs, microservices.

6. Monitoring & Feedback Layer

- Tracks performance in real-world use.
- Retrains model with new data to maintain accuracy.

Example Flow

- **Input:** Customer purchase history.
- **Process:** Data preprocessing → Feature engineering → Train ML model (e.g., Decision Tree).
- **Output:** Predict customer churn → Integrated into CRM system.
- **Monitoring:** Continuous feedback loop to improve predictions.

b. Describe any four uninformed search strategies. Compare them with time complexity, space complexity, optimality and completeness.

Definition:

Uninformed search strategies (also called blind search) explore the search space **without using heuristics**. They rely only on the problem definition (initial state, actions, goal state).

Four Main Strategies

1. Breadth-First Search (BFS)

- Explores nodes level by level (shallowest first).
- Uses a **queue (FIFO)**.
- Guarantees shortest path if step costs are equal.

2. Depth-First Search (DFS)

- Explores deep into one branch before backtracking.
- Uses a **stack (LIFO)** or recursion.
- Memory-efficient but may get stuck in infinite paths.

3. Uniform Cost Search (UCS)

- Expands the node with the **lowest path cost**.
- Uses a **priority queue** ordered by cumulative cost $g(n)$.
- Guarantees optimal solution when costs vary.

4. Depth-Limited / Iterative Deepening DFS

- DFS with a depth limit to avoid infinite loops.
- Iterative deepening combines DFS depth efficiency with BFS completeness.

Comparison Table				
Algorithm	Time Complexity	Space Complexity	Completeness	Optimality
BFS	$O(b^d)$	$O(b^d)$	Yes	Yes (if cost same)
DFS	$O(b^m)$	$O(bm)$	No (can fail in infinite depth)	No
UCS	$O(b^{1+\lceil C^*/\epsilon \rceil})$	Same as time	Yes	Yes
IDDFS	$O(b^d)$	$O(bd)$	Yes	Yes (if cost same)

Where:

- b = branching factor
- d = depth of goal
- m = maximum depth
- C^* = optimal cost
- ϵ = minimum step cost

AIDS 2024 December PYQ Answers

Q1. [5 Marks each]

- a) What are the different issues in ML algorithms?
- b) Compare barplot and histogram.
- c) Define and state effects of overfitting and underfitting.
- d) Explain learning agents with example.

Q2. [10 Marks each]

- a) What do you mean by covariance and correlation? Explain the range of coefficients of correlation and covariance. Calculate COV (Observed Value1, Observed Value2) and CORR (Observed Value1, Observed Value2) for following data. How do you interpret these values?

Experiment No.	1	2	3	4	5	6	7	8	9	10
Observed Value 1	20	10	30	15	45	40	30	35	45	30
Observed value 2	20	15	30	10	40	45	20	35	40	25

- b) Compare Z-Test, T-Test and ANOVA in detail.

Q3. [10 Marks each]

- a) What is SVM? Explain its significance in ML and compare it with logistic regression.
- b) What are the different univariate plots in EDA? Explain them in detail.

Q4. [10 Marks each]

- a) Differentiate between data scientists, big data professionals and data analysts.

b) What is linear regression? Explain its significance in ML and compare it with logistic regression.

Q5. [10 Marks each]

a) Describe steps for developing ML applications with a labeled diagram.

b) What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.

Q6. [5 Marks each]

Write Short Note on (Any four):

a) CNF

b) Alpha Beta Pruning

c) A* algorithm

d) Skolemization

e) Rules in Propositional Logic

Q1. [5 Marks each] - Answers

b. Compare barplot and histogram.

Basis	Bar Plot	Histogram
Definition	Represents categorical data using bars	Represents frequency distribution of continuous data
Data Type	Categorical (discrete categories)	Continuous numerical data
Bars	Bars are separated (gaps present)	Bars are touching (no gaps)
Purpose	Compare values across categories	Show data distribution and frequency
X-axis	Categories (e.g., names, labels)	Intervals/bins of values
Order	No fixed order required	Ordered (numerical intervals)

Basis	Bar Plot	Histogram
Example	Number of students in each class	Distribution of student marks

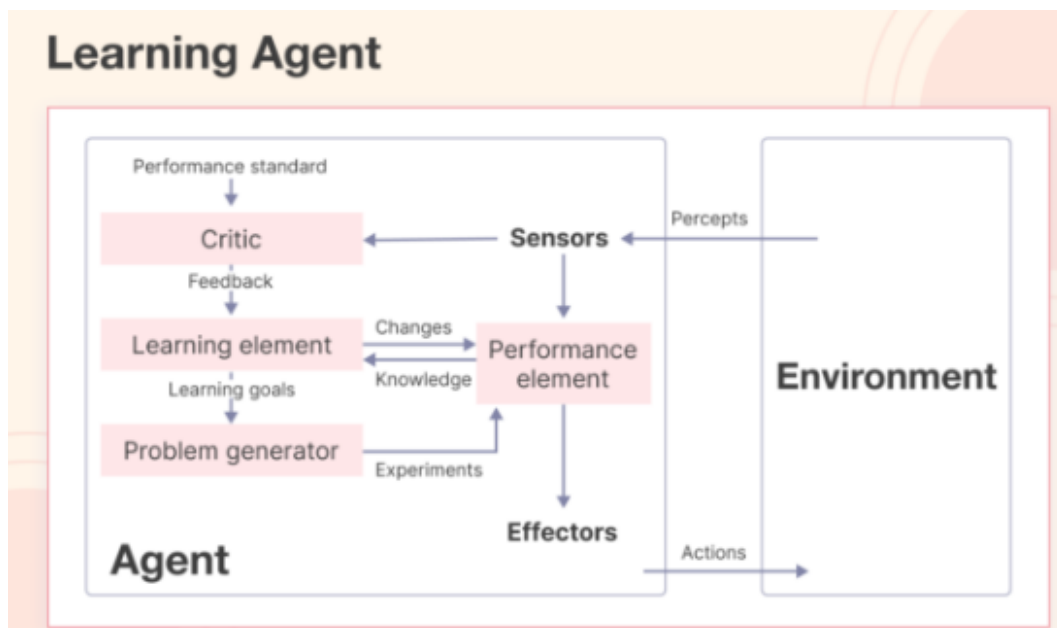
d. Explain learning agents with example.

Definition:

A **Learning Agent** is an AI agent that improves its performance over time by learning from past experiences and adapting to new situations.

Key Components:

1. **Learning Element:** Responsible for making improvements based on feedback.
2. **Performance Element:** Chooses actions based on current knowledge.
3. **Critic:** Evaluates the agent's performance and provides feedback.
4. **Problem Generator:** Suggests exploratory actions to discover new knowledge.



Working:

- The agent interacts with the environment.
- The **performance element** selects actions.

- The **critic** assesses outcomes against performance measures.
- The **learning element** updates knowledge or strategies.
- The **problem generator** encourages exploration to avoid stagnation.

Example:

- **Chess-playing Agent:**
 - Initially plays with basic strategies.
 - Learns from wins/losses (feedback).
 - Updates its strategy (learning element).
 - Over time, improves performance and plays stronger games.

Q3. [10 Marks each] - Answers

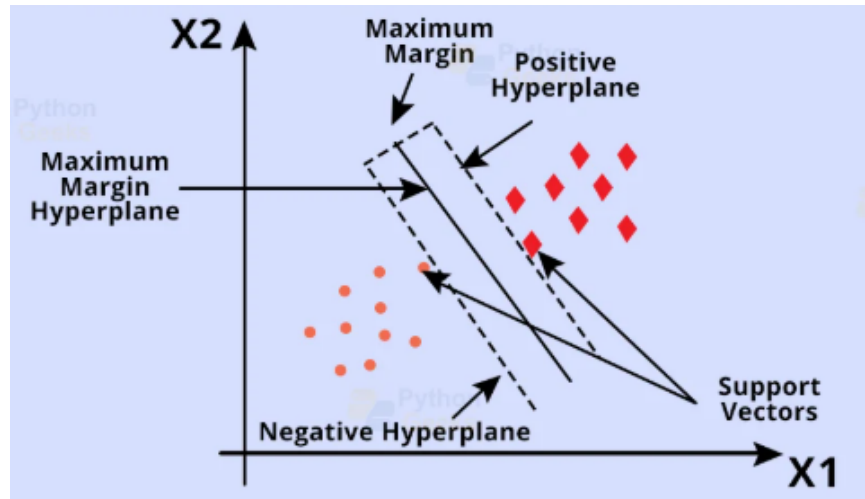
a. What is SVM? Explain its significance in ML and compare it with logistic regression.

i) Support Vector Machine (SVM)

SVM is a supervised learning algorithm used mainly for **classification**. It finds the **optimal hyperplane** that separates data points of different classes with maximum margin.

Key Concepts

- **Hyperplane:** Decision boundary separating classes.
- **Margin:** Distance between hyperplane and nearest data points.
- **Support Vectors:** Data points closest to the hyperplane.
- **Kernel Trick:** Transforms data into higher dimensions (linear, polynomial, RBF).



Working

1. Plot data points in feature space.
2. Find a hyperplane that separates classes.
3. Maximize margin between classes.
4. Use kernel if data is not linearly separable.

Example

- Email classification: Spam vs Not Spam.

Significance of SVM in ML

- Effective in **high-dimensional data** (e.g., text classification).
- Works well with **small and medium datasets**.
- Provides **clear margin of separation**, improving generalization.
- Robust to overfitting (especially with proper kernel and regularization).
- Can handle **linear and non-linear problems** using kernels.

Aspect	SVM	Logistic Regression
Type of Problem	Classification (also regression)	Classification

Aspect	SVM	Logistic Regression
Approach	Maximizes margin between classes	Uses probability (sigmoid function)
Decision Boundary	Based on support vectors	Based on probability threshold
Output	Class label	Probability (0 to 1)
Handling Non-linearity	Uses kernel trick	Requires feature transformation
Performance	Better for complex, high-dimensional data	Better for simple, linearly separable data
Interpretability	Less interpretable	More interpretable
Computation	Slower for large datasets	Faster and scalable
Example	Image classification	Spam detection

Q6. [5 Marks each] - Answers

a. Write Short Note on: CNF

Definition:

CNF (Conjunctive Normal Form) is a standard form of logical expression where a formula is written as a **conjunction (AND) of clauses**, and each clause is a **disjunction (OR) of literals**.

Key Points:

- Structure: $(A \vee B) \wedge (\neg C \vee D)$
- Each clause contains **literals** (variables or their negations).
- Used in **automated theorem proving and resolution**.
- Simplifies logical expressions for **efficient inference**.
- Conversion steps include:
 - Remove implications ($\rightarrow$)
 - Move NOT inward (using De Morgan's laws)

- Standardize variables
- Apply Skolemization
- Distribute OR over AND
- Essential for **resolution-based reasoning systems**.

Example:

Expression: $(A \rightarrow B)$

$\rightarrow (\neg A \vee B)$ (in CNF form)

b. Write Short Note on: Alpha Beta Pruning

Definition:

Alpha-Beta pruning is an optimization technique for the **Min-Max algorithm** used in game tree search. It eliminates branches that cannot affect the final decision, thereby reducing computation.

Key Concepts

- **Alpha (α):** Best value that the maximizer can guarantee so far.
- **Beta (β):** Best value that the minimizer can guarantee so far.
- **Pruning Condition:** If at any point $\alpha \geq \beta$, further exploration of that branch is unnecessary.

Working

1. Start with the root node (MAX).
2. Explore child nodes, updating α and β values.
3. If a branch cannot improve the current decision, prune it (skip evaluation).
4. Continue until the optimal move is found with fewer expansions.

Example

- In a game tree, if MIN already has a better option than what MAX can achieve in a branch, that branch is **cut off**.
- Saves time without changing the final decision.

c. Write Short Note on: A* Algorithm

Definition:

A* is an **informed search algorithm** used for **pathfinding and graph traversal**. It combines the actual cost from the start node ($g(n)$) with an estimated cost to the goal ($h(n)$) using the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$: Cost from start to current node
- $h(n)$: Heuristic (estimated cost to goal)
- $f(n)$: Total estimated cost

Key Features

- Uses both **path cost (g)** and **heuristic estimate (h)**.
- Expands the node with the **lowest** $f(n)$ value.
- Guarantees **optimality** if the heuristic is admissible (never overestimates).
- More efficient than uninformed search methods like BFS or DFS.

Example

- In a grid pathfinding problem:
 - $g(n)$: distance traveled so far.
 - $h(n)$: Manhattan distance to the goal.
 - A* selects the path with the smallest $f(n)$, ensuring the shortest route.

d. Write Short Note on: Skolemization

Definition:

Skolemization is a process in **predicate logic** used to eliminate **existential quantifiers ($\exists$)** by replacing them with **Skolem constants or Skolem functions**. It

transforms formulas into a quantifier-free form, making them suitable for **automated reasoning and resolution proofs**.

Skolem Constant

- Used when the existential quantifier does **not depend** on any universally quantified variable.
- Replaced with a **new constant symbol**.
- Example:
 - $\exists x P(x) \rightarrow P(c)$, where **c** is a Skolem constant.

Skolem Function

- Used when the existential quantifier **depends on universally quantified variables**.
- Replaced with a **function of those variables**.
- Example:
 - $\forall y \exists x P(x, y) \rightarrow \forall y P(f(y), y)$, where **f(y)** is a Skolem function.

e. Write Short Note on: Rules in Propositional Logic

Definition:

Rules in **Propositional Logic** are inference principles that allow us to derive new propositions from existing ones. They form the basis of **logical reasoning** in AI and mathematics.

Key Rules

1. Modus Ponens (Implication Rule):

- If $P \rightarrow Q$ and P are true, then Q is true.
- Example: If it rains, the ground gets wet. It rains $\rightarrow$ Ground is wet.

2. Modus Tollens:

- If $P \rightarrow Q$ and $\neg Q$, then $\neg P$.
- Example: If it rains, the ground gets wet. Ground not wet $\rightarrow$ It didn't rain.

3. Disjunctive Syllogism:

- If $P \vee Q$ and $\neg P$, then Q .

4. Resolution Rule:

- From $(P \vee Q)$ and $(\neg P \vee R)$, infer $(Q \vee R)$.
- Widely used in automated theorem proving.

AIDS 2023 May PYQ Answers

Q1. [5 Marks each]

- a) Differentiate Between Forward and Backward chaining.
- b) Compare different search techniques based on their time complexities.
- c) What is a histogram? Can we perform univariate graphical analysis using histogram?
- d) Explain various measures of the central tendencies of a statistical distribution.
- e) State PEAS of automated taxi driver.
- f) What are the different ways of knowledge representation?

Q2. [10 Marks each]

- a) Can 1 liter water be measured using 10 liter and 4 liter jug? Justify.
- b) Compare Linear Regression Vs Logistics Regression with suitable diagrams and formulas.

Q3. [10 Marks each]

- a) State A* algorithm and explain with example how searching algorithm helps A* in finding the goal with optimal path.
- b) With respect to Quantitative data analysis explain following:
 - i. Measure of central tendencies
 - ii. Measure of spread
 - iii. Skewness and Kurtosis

Q4. [10 Marks each]

a) Resolution Problem:

1. Marcus was a man.
2. Marcus was a Pompeian.
3. All Pompeians were Romans.
4. Caesar was a ruler.
5. All Pompeians were either loyal to Caesar or hated him.
6. Every one is loyal to someone.
7. People only try to assassinate rulers they are not loyal to.

8. Marcus tried to assassinate Caesar.

Was Marcus loyal to Caesar? Solve using resolution.

b) In detail, explain steps in the Data Science Project.

Q5. [10 Marks each]

a) What are the different types of Machine Learning algorithms? Give example of proper example.

b) Can min-max be used for team games? Draw sample trees for 2 and 3 teams.

Q6. [10 Marks each]

a) Consider you are performing ML for predicting housing prices you have trained three models and following data summarizes the predicted house price (in Lakh Rs) by each model for 5 different trial runs:

Model Code	House Price Predicted (Lakh Rs)				
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
A	3.5	3.4	3.8	3.5	3.4
B	3.9	3.8	3.7	3.9	3.6
C	3.5	3.3	3.6	3.5	3.8

Perform One way ANOVA F Test on this data and comment on whether the mean house price predicted by models A, B, C are same with level of significance 0.05. (Use of F Table is allowed).

b) What are the rules of conversion from predicate to CNF? Explain each rule with proper example.

Q1. [5 Marks each] - Answers

a. Differentiate Between Forward and Backward chaining.

Basis	Forward Chaining	Backward Chaining
Approach	Data-driven	Goal-driven
Starting Point	Starts from known facts	Starts from goal/query
Direction	Moves from facts → conclusion	Moves from goal → facts

Basis	Forward Chaining	Backward Chaining
Process	Applies rules to generate new facts	Breaks goal into sub-goals
Termination	Stops when goal is reached or no rules apply	Stops when goal is proven or fails
Efficiency	Less efficient if many irrelevant facts	More efficient for specific goals
Use Case	Prediction, monitoring systems	Diagnosis, query answering
Example	$A \rightarrow B \rightarrow C$ (derive step by step)	To prove $C \rightarrow$ check $B \rightarrow$ check A

b. Compare different search techniques based on their time complexities.

Search Technique	Time Complexity	Description
Breadth First Search (BFS)	$O(b^d)$	Explores level by level; guarantees optimal solution
Depth First Search (DFS)	$O(b^m)$	Explores deepest nodes first; may not be optimal
Uniform Cost Search (UCS)	$O(b^{(1 + \lceil C^*/\epsilon \rceil)})$	Expands lowest cost path first
Greedy Best-First Search	$O(b^m)$	Uses heuristic; fast but not always optimal
A* Search	$O(b^d)$ (worst case)	Uses cost + heuristic; optimal if heuristic is admissible

Where:

- **b** = branching factor
- **d** = depth of solution
- **m** = maximum depth
- **C*** = optimal solution cost
- **ε** = minimum step cost

c. What is a histogram? Can we perform univariate graphical analysis using histogram?

Histogram

- A **Histogram** is a graphical representation of the **frequency distribution** of continuous numerical data.
- Data is divided into **intervals (bins)**, and the height of each bar shows how many values fall within that range.

- Bars are **adjacent (no gaps)** since the data is continuous.

Use in Univariate Graphical Analysis

- Yes, histograms are widely used for **univariate graphical analysis**, as they show the distribution of a **single variable** clearly.
- They help in identifying **spread, skewness, central tendency, and outliers**.
- Example: A histogram of exam scores shows how many students fall into each score range.

d. Explain various measures of the central tendencies of a statistical distribution.

Definition:

Measures of central tendency are statistical values that represent the **center or typical value** of a dataset.

Key Points:

- **Mean (Arithmetic Average):**
Sum of all values divided by total number of values.
Sensitive to extreme values (outliers).
- **Median:**
Middle value when data is arranged in order.
Not affected by outliers; good for skewed data.
- **Mode:**
Most frequently occurring value in the dataset.
Useful for categorical data.

Example:

Data: 2, 4, 4, 6, 8

- Mean = 4.8
- Median = 4
- Mode = 4

e. State PEAS of automated taxi driver.

Component	Automated Taxi Driver Description
Performance Measure	Safe, fast, and comfortable transport; obey traffic rules; minimize travel time; maximize passenger satisfaction.
Environment	Roads, traffic signals, pedestrians, other vehicles, weather conditions, passengers.
Actuators	Steering wheel, accelerator, brake, horn, indicators, doors.
Sensors	Cameras, GPS, speedometer, lidar/radar, microphones, proximity sensors.

f. What are the different ways of knowledge representation?

Definition:

Knowledge representation in AI refers to the methods used to encode information about the world so that a computer system can reason and make decisions.

Different Ways:

1. Logical Representation:

- Uses formal logic (propositional, predicate logic).
- Example: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$.

2. Semantic Networks:

- Represents knowledge as **nodes (concepts)** and **edges (relationships)**.
- Example: Dog $\rightarrow$ is-a $\rightarrow$ Animal.

3. Frames:

- Data structures with slots and values to represent stereotypical situations.
- Example: Frame for "Car" with slots: {Make, Model, Color}.

4. Production Rules:

- IF-THEN rules used in expert systems.
- Example: IF fever $\wedge$ cough THEN flu.

5. Ontologies:

- Structured representation of concepts and relationships in a domain.
- Example: Medical ontology linking diseases, symptoms, and treatments.

Q2. [10 Marks each] - Answers

a. Can 1 liter water be measured using 10 liter and 4 liter jug? Justify.

This is a **Water Jug Problem**, where we check if a desired quantity can be measured using given jug capacities.

Key Points:

- Given jugs: **10 L and 4 L**, target = **1 L**.
- Problem can be solved only if target is a multiple of **GCD of jug sizes**.
- $\text{GCD}(10, 4) = 2$.
- Possible measurable quantities = multiples of 2 $\rightarrow (2, 4, 6, 8, 10\dots)$.
- **1 liter is not a multiple of 2**, so it cannot be obtained.
- Therefore, it is **not possible** to measure exactly 1 liter using these jugs.

Example:

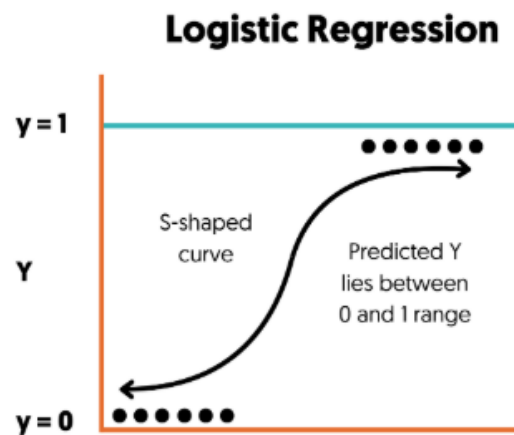
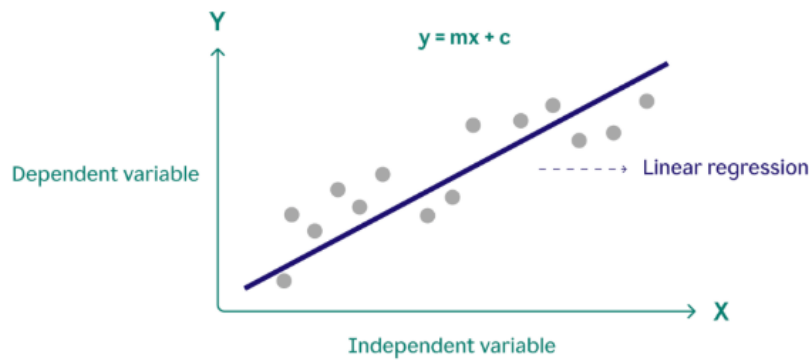
We can measure 2 L, 4 L, 6 L, etc., but **never exactly 1 L**.

b. Compare Linear Regression Vs Logistics Regression with suitable diagrams and formulas.

Basis	Linear Regression	Logistic Regression
Purpose	Predict continuous values	Predict categorical outcomes (0/1)
Output	Continuous (e.g., price, marks)	Probability (0 to 1)
Type of Problem	Regression	Classification
Function Used	Linear function	Sigmoid (logistic) function
Equation	$y = mx + c$	$p = 1 / (1 + e^{(-z)})$
Output Range	$(-\infty \text{ to } +\infty)$	(0 to 1)
Decision Boundary	Straight line	S-shaped curve
Error Measurement	Mean Squared Error (MSE)	Log Loss (Cross-Entropy)
Example	Predict house price	Predict spam (yes/no)

Formulas:

- **Linear Regression:**
$$y = \beta_0 + \beta_1 x$$
- **Logistic Regression:**
$$p = 1 / (1 + e^{-(\beta_0 + \beta_1 x)})$$



Q3. [10 Marks each] - Answers

a. State A* algorithm and explain with example how searching algorithm helps A* in finding the goal with optimal path.

Definition:

A* is an **informed search algorithm** used for **pathfinding and graph traversal**. It combines the actual cost from the start node ($g(n)$) with an estimated cost to the goal ($h(n)$) using the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$: Cost from start to current node

- $h(n)$: Heuristic (estimated cost to goal)
- $f(n)$: Total estimated cost

Key Features

- Uses both **path cost (g)** and **heuristic estimate (h)**.
- Expands the node with the **lowest** $f(n)$ value.
- Guarantees **optimality** if the heuristic is admissible (never overestimates).
- More efficient than uninformed search methods like BFS or DFS.

Example (Path Finding):

Consider nodes:

Start $\rightarrow$ A $\rightarrow$ Goal

- Cost from Start to A = 2 $\rightarrow$ **$g(A) = 2$**
- Heuristic from A to Goal = 3 $\rightarrow$ **$h(A) = 3$**
- So, **$f(A) = 2 + 3 = 5$**

Now compare another path:

Start $\rightarrow$ B $\rightarrow$ Goal

- $g(B) = 4$, $h(B) = 1 \rightarrow f(B) = 5$

A* will choose node with lowest **$f(n)$** (tie $\rightarrow$ any, but still optimal).

How A* finds optimal path:

- Evaluates both **actual cost (g)** and **estimated future cost (h)**
- Avoids paths that are too long or misleading
- Expands only **most promising nodes first**
- Ensures shortest path is found when **$h(n)$ is accurate**

b. With respect to Quantitative data analysis explain following:

- i. Measure of central tendencies
- ii. Measure of spread
- iii. Skewness and Kurtosis

Definition:

Quantitative data analysis uses statistical measures to **summarize, describe, and understand numerical data**.

i. Measure of Central Tendencies

- Indicates the **central or typical value** of data.
- **Mean:** Average of all values.
- **Median:** Middle value when data is sorted.
- **Mode:** Most frequent value.
- Helps understand the **overall trend of data**.
- Median is preferred when data has **outliers**.

Example:

Marks: 2, 4, 6 → Mean = 4

ii. Measure of Spread (Dispersion)

- Shows how data is **spread or dispersed** around the center.
- **Range:** Difference between max and min values.
- **Variance:** Average of squared deviations from mean.
- **Standard Deviation:** Square root of variance.
- **Interquartile Range (IQR):** Spread of middle 50% data.
- Helps understand **consistency and variability**.

Example:

Low spread → values close together; High spread → values widely spread

iii. Skewness and Kurtosis

- **Skewness:** Measures **asymmetry** of data distribution.
 - Positive skew → tail on right
 - Negative skew → tail on left
- **Kurtosis:** Measures **peakedness or flatness** of distribution.
 - High kurtosis → sharp peak (leptokurtic)
 - Low kurtosis → flat distribution (platykurtic)
- Helps understand **shape of data distribution**.

Example:

Income data is often **positively skewed** (few people earn very high incomes).

Q4. [10 Marks each] - Answers

a. Resolution Problem:

1. Marcus was a man.
2. Marcus was a Pompeian.
3. All Pompeians were Romans.
4. Caesar was a ruler.
5. All Pompeians were either loyal to Caesar or hated him.
6. Every one is loyal to someone.
7. People only try to assassinate rulers they are not loyal to.
8. Marcus tried to assassinate Caesar.

Was Marcus loyal to Caesar? Solve using resolution.

Step 1: Convert Statements into Predicate Logic

Let:

- $\text{Man}(x), \text{Pompeian}(x), \text{Roman}(x), \text{Ruler}(x)$
- $\text{Loyal}(x,y), \text{Hates}(x,y), \text{TryAssassinate}(x,y)$

Given:

1. Marcus is a man
 $\rightarrow \text{Man}(\text{Marcus})$
2. Marcus is a Pompeian
 $\rightarrow \text{Pompeian}(\text{Marcus})$
3. All Pompeians are Romans
 $\rightarrow \neg \text{Pompeian}(x) \vee \text{Roman}(x)$
4. Caesar is a ruler
 $\rightarrow \text{Ruler}(\text{Caesar})$
5. All Pompeians are loyal or hate Caesar
 $\rightarrow \neg \text{Pompeian}(x) \vee \text{Loyal}(x, \text{Caesar}) \vee \text{Hates}(x, \text{Caesar})$
6. Everyone is loyal to someone
 $\rightarrow \exists y \text{Loyal}(x,y)$ (not directly needed)
7. People try to assassinate only rulers they are not loyal to
 $\rightarrow \neg \text{TryAssassinate}(x,y) \vee \neg \text{Ruler}(y) \vee \neg \text{Loyal}(x,y)$

8. Marcus tried to assassinate Caesar
→ TryAssassinate(Marcus, Caesar)
-

Step 2: Convert to CNF Clauses

1. Pompeian(Marcus)
 2. $\neg \text{Pompeian}(x) \vee \text{Roman}(x)$
 3. Ruler(Caesar)
 4. $\neg \text{Pompeian}(x) \vee \text{Loyal}(x, \text{Caesar}) \vee \text{Hates}(x, \text{Caesar})$
 5. $\neg \text{TryAssassinate}(x, y) \vee \neg \text{Ruler}(y) \vee \neg \text{Loyal}(x, y)$
 6. TryAssassinate(Marcus, Caesar)
-

Step 3: Resolution Process

From (5) and (6):

- $\neg \text{TryAssassinate}(x, y) \vee \neg \text{Ruler}(y) \vee \neg \text{Loyal}(x, y)$
- TryAssassinate(Marcus, Caesar)

Substitute →

$\Rightarrow \neg \text{Ruler}(\text{Caesar}) \vee \neg \text{Loyal}(\text{Marcus}, \text{Caesar})$

From above and (3):

- Ruler(Caesar)

$\Rightarrow \neg \text{Loyal}(\text{Marcus}, \text{Caesar})$

Step 4: Check with Pompeian Rule

From (4) and Pompeian(Marcus):

$\Rightarrow \text{Loyal}(\text{Marcus}, \text{Caesar}) \vee \text{Hates}(\text{Marcus}, \text{Caesar})$

But we already proved:

$\Rightarrow \neg \text{Loyal}(\text{Marcus}, \text{Caesar})$

So:

$\Rightarrow \text{Hates}(\text{Marcus}, \text{Caesar})$

b. In detail, explain steps in the Data Science Project.

Steps in a Data Science Project

1. Problem Definition

- Clearly define the business or research problem.
- Example: Predict customer churn for a telecom company.

2. Data Collection

- Gather relevant data from sources such as databases, APIs, sensors, or surveys.
- Example: Customer demographics, call records, billing history.

3. Data Cleaning & Preprocessing

- Handle missing values, remove duplicates, correct inconsistencies.
- Normalize or standardize data for consistency.

4. Exploratory Data Analysis (EDA)

- Use statistical methods and visualization to understand distributions, relationships, and anomalies.
- Example: Histograms, scatter plots, correlation matrices.

5. Feature Engineering & Selection

- Create new features or select the most relevant ones to improve model performance.
- Example: Deriving "average call duration" from raw call logs.

6. Model Building

- Choose appropriate algorithms (Regression, Classification, Clustering, etc.).
- Train models using training datasets.

7. Model Evaluation

- Assess performance using metrics like accuracy, precision, recall, F1-score, RMSE.
- Perform cross-validation to ensure robustness.

8. Deployment

- Integrate the model into production systems (e.g., web app, API).
- Example: Real-time churn prediction integrated into CRM software.

9. Monitoring & Maintenance

- Continuously track model performance.
- Update or retrain models when data distribution changes (concept drift).

Q5. [10 Marks each] - Answers

a. What are the different types of Machine Learning algorithms? Give example of proper example.

Definition

Machine Learning algorithms are classified based on how they learn from data. The main types are **Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning**.

Types of Machine Learning Algorithms with Examples

1. Supervised Learning

- **Definition:** Learns from labeled datasets (input-output pairs).
- **Goal:** Predict outcomes for new unseen data.
- **Examples:**
 - **Linear Regression** → Predicting house prices based on area.
 - **Logistic Regression** → Spam email detection (spam/not spam).
 - **Decision Trees** → Classifying whether a customer will churn.
 - **Support Vector Machines (SVM)** → Handwritten digit recognition.
 - **Naive Bayes** → Text classification (e.g., sentiment analysis).
 - **Random Forests** → Loan approval prediction.

2. Unsupervised Learning

- **Definition:** Works with unlabeled data to find hidden structures or groupings.
- **Goal:** Discover patterns without predefined labels.
- **Examples:**
 - **K-Means Clustering** → Customer segmentation in marketing.
 - **Hierarchical Clustering** → Grouping genes with similar expression patterns.
 - **Association Rule Learning (Apriori, FP-Growth)** → Market basket analysis (e.g., "If a customer buys bread, they also buy butter").

3. Reinforcement Learning

- **Definition:** Agent learns by interacting with an environment, receiving rewards or penalties.
- **Goal:** Maximize cumulative reward through trial and error.
- **Examples:**
 - **Q-Learning** → Robot navigation in a maze.
 - **Deep Reinforcement Learning (DQN)** → Playing video games like Atari.

- **Policy Gradient Methods** → Self-driving cars decision-making.
- **AlphaGo** → AI beating humans in the game of Go.

b. Can min-max be used for team games? Draw sample trees for 2 and 3 teams.

Definition

Minimax is a decision-making algorithm used in adversarial games (like chess) where two players have **opposite goals**. It assumes one player maximizes the score while the other minimizes it.

Can Minimax be used for Team Games?

- **Standard Minimax is designed for 2-player zero-sum games only.**
- It works when:
 - One player = **MAX (maximize score)**
 - Other player = **MIN (minimize score)**
- In **team/multi-player games (3 or more players)**:
 - Players may not have strictly opposite goals.
 - There are **multiple utilities**, not a single value.
- Hence, Minimax is **not directly suitable**.
- Instead, extensions like **Maxⁿ algorithm** are used for multi-player games.

Q6. [10 Marks each] - Answers

a. Consider you are performing ML for predicting housing prices you have trained three models and following data summarizes the predicted house price (in Lakh Rs) by each model for 5 different trial runs. Perform One way ANOVA F Test on this data and comment on whether the mean house price predicted by models A, B, C are same with level of significance 0.05. (Use of F Table is allowed).

Given data:

Model	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
A	3.5	3.4	3.8	3.5	3.4
B	3.9	3.8	3.7	3.9	3.6

Model	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
C	3.5	3.3	3.6	3.5	3.8

Level of significance:

$$\alpha = 0.05$$

Step 1: State Hypotheses

Null Hypothesis (H_0)

Mean house prices predicted by models A, B, and C are same.

$$\mu_A = \mu_B = \mu_C$$

Alternate Hypothesis (H_1)

At least one model has different mean prediction.

Step 2: Calculate Group Means

Mean of Model A

$$\begin{aligned}\bar{A} &= (3.5 + 3.4 + 3.8 + 3.5 + 3.4)/5 \\ &= 17.6/5 \\ &= 3.52\end{aligned}$$

Mean of Model B

$$\begin{aligned}\bar{B} &= (3.9 + 3.8 + 3.7 + 3.9 + 3.6)/5 \\ &= 18.9/5 \\ &= 3.78\end{aligned}$$

Mean of Model C

$$\begin{aligned}\bar{C} &= (3.5 + 3.3 + 3.6 + 3.5 + 3.8)/5 \\ &= 17.7/5 \\ &= 3.54\end{aligned}$$

Step 3: Calculate Grand Mean

$$\begin{aligned}\text{Grand Mean} &= (17.6 + 18.9 + 17.7)/15 \\ &= 54.2/15 \\ &= 3.613\end{aligned}$$

Step 4: Calculate Sum of Squares

Sum of Squares Between Groups (SSB)

$$SSB = \sum n(\bar{X}_i - \bar{X})^2$$

$$SSB = 5(3.52 - 3.613)^2 + 5(3.78 - 3.613)^2 + 5(3.54 - 3.613)^2$$

$$SSB = 0.0433 + 0.1394 + 0.0266$$

$$SSB \approx 0.2093$$

Sum of Squares Within Groups (SSW)

$$\text{For A} = 0.108$$

$$\text{For B} = 0.068$$

$$\text{For C} = 0.148$$

$$\begin{aligned}SSW &= 0.108 + 0.068 + 0.148 \\ &= 0.324\end{aligned}$$

Step 5: Degrees of Freedom

$$df_{\text{between}} = k - 1 = 3 - 1 = 2$$

$$df_{\text{within}} = N - k = 15 - 3 = 12$$

Where:

- k = number of groups
- N = total observations

Step 6: Calculate Mean Squares

Mean Square Between (MSB)

$$\begin{aligned} \text{MSB} &= \text{SSB} / \text{dfbetween} \\ &= 0.2093 / 2 \\ &= 0.10465 \end{aligned}$$

Mean Square Within (MSW)

$$\begin{aligned} \text{MSW} &= \text{SSW} / \text{dfwithin} \\ &= 0.324 / 12 \\ &= 0.027 \end{aligned}$$

Step 7: Calculate F-ratio

$$F = \frac{\text{MSB}}{\text{MSW}}$$

$$\begin{aligned} F &= 0.10465 / 0.027 \\ &\approx 3.88 \end{aligned}$$

Step 8: Compare with F-table Value

From F-table at:

- $\alpha = 0.05$
- $\text{df1} = 2$
- $\text{df2} = 12$

$$F_{\text{critical}} \approx 3.89$$

Comparison:

$$F_{\text{calculated}} = 3.88$$

$$F_{\text{critical}} = 3.89$$

$$3.88 < 3.89$$

Decision

Since calculated F-value is less than critical F-value:

Fail to reject Null Hypothesis (H_0)

Interpretation

- There is no statistically significant difference between the mean house prices predicted by Models A, B, and C.
- The observed variation may be due to random chance.

Conclusion

Using One-Way ANOVA at significance level 0.05:

- The mean predicted house prices of Models A, B, and C are statistically same.
- ANOVA fails to reject the Null Hypothesis.

b. What are the rules of conversion from predicate to CNF? Explain each rule with proper example.

Rules for Conversion to CNF (with Examples)

1. Eliminate Biconditionals and Implications

- Rule: Replace
 - $P \leftrightarrow Q$ with $(P \rightarrow Q) \wedge (Q \rightarrow P)$.
 - $P \rightarrow Q$ with $\neg P \vee Q$.
- Example:
 - $\forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$
 - $\rightarrow \forall x(\neg \text{Human}(x) \vee \text{Mortal}(x)).$

2. Move Negations Inward (Negation Normal Form)

- Rule: Apply **De Morgan's laws** and quantifier negation rules:
 - $\neg(\forall x P(x)) \equiv \exists x \neg P(x).$

- $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$.

- Example:

$$\neg \forall x \text{Loves}(x, \text{IceCream})$$

$$\rightarrow \exists x \neg \text{Loves}(x, \text{IceCream}).$$

3. Standardize Variables Apart

- Rule: Rename bound variables so each quantifier uses a unique variable.

- Example:

$$\forall x P(x) \wedge \forall x Q(x)$$

$$\rightarrow \forall x P(x) \wedge \forall y Q(y).$$

4. Skolemization (Remove Existential Quantifiers)

- Rule: Replace existential quantifiers with **Skolem constants/functions**.

- Example:

$$\forall x \exists y \text{Loves}(x, y)$$

$$\rightarrow \forall x \text{Loves}(x, f(x)), \text{ where } f(x) \text{ is a Skolem function.}$$

5. Drop Universal Quantifiers

- Rule: Once Skolemization is done, all variables are implicitly universally quantified.

- Example:

$$\forall x \text{Mortal}(x)$$

$$\rightarrow \text{Mortal}(x).$$

6. Distribute OR over AND

- Rule: Ensure formula is a conjunction of disjunctions.

- Example:

$$(P \vee (Q \wedge R))$$

$$\rightarrow (P \vee Q) \wedge (P \vee R).$$

AIDS 2023 December PYQ Answers

Q1. [5 Marks each]

- a) Write comparison between Business Intelligence and Data Science.
- b) What is rational agent? Explain with diagram.
- c) Explain what role is played by Correlation and Covariance in EDA?
- d) What are heuristic functions? Where are they used?
- e) What is Unification? Give example.
- f) What is Skolemization? Explain Skolem constant and Skolem function.

Q2. [10 Marks each]

- a) Write a detailed note on hypothesis testing. What are type I and type II errors?
- b) The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. **Prove that Col. West is a criminal!**

Q3. [10 Marks each]

- a) Explain uniform cost search and best first search in detail with examples and compare.
- b) Explain various stages in the Data analytics lifecycle.

Q4. [10 Marks each]

- a) Explain SVM in detail.
- b) Describe PEAS and also write down the PEAS representations for Medical diagnosis system.

Q5. [10 Marks each]

- a) Write in detail issues in machine learning.
- b) Elaborate in detail the steps in developing a Machine Learning application with architectural diagram.

Q6. [10 Marks each]

- a) What are the different planning techniques? Explain with example.
- b) What do you mean by covariance and correlation? Explain the range of coefficients of correlation and covariance. Calculate COV (Observed Value1, Observed Value2) and CORR (Observed Value1, Observed Value2) for following data. How do you interpret these values?

Experiment No	Observed Value1	Observed Value2	Experiment No	Observed Value1	Observed Value2
1	38	20	9	80	9
2	62	15	10	32	22
3	22	30	11	51	20
4	38	21	12	56	19
5	45	18	13	21	28
6	69	12	14	34	23
7	75	14	15	76	14
8	38	28			

Q1. [5 Marks each] - Answers

a. Write comparison between Business Intelligence and Data Science.

Basis	Business Intelligence (BI)	Data Science
Definition	Focuses on analyzing past data to support business decisions	Uses data to build models and predict future outcomes
Goal	Reporting, monitoring, and decision support	Prediction, pattern discovery, and automation
Data Type	Mostly structured data	Structured + unstructured data
Techniques Used	SQL, OLAP, dashboards, reporting tools	Machine learning, statistics, programming
Approach	Descriptive and diagnostic analysis	Predictive and prescriptive analysis
Output	Reports, charts, dashboards	Models, predictions, insights
Example	Sales report dashboard	Predicting customer churn

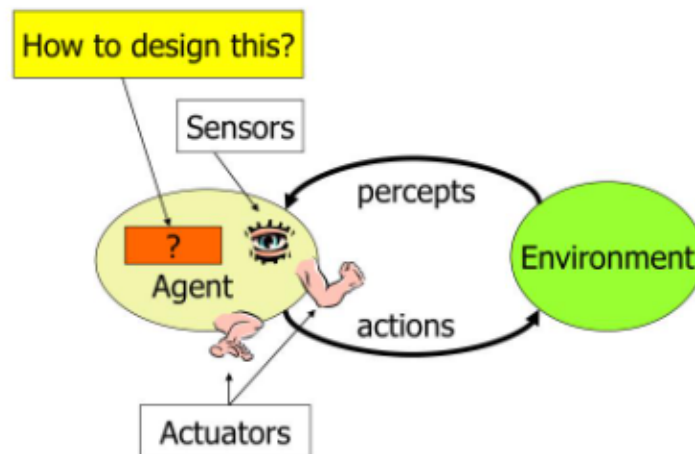
b. What is rational agent? Explain with diagram.

Definition:

A **Rational Agent** is an AI agent that always chooses actions expected to maximize its performance measure, based on its percepts, knowledge, and available actions.

Key Points:

- **Rationality:** Depends on performance measure, percept sequence, knowledge of environment, and available actions.
- **Goal:** Select actions that lead to the best possible outcome.
- **Adaptability:** Can adjust decisions based on changing environment.
- **Example:** An autonomous taxi that chooses the safest and fastest route considering traffic and weather conditions.



c. Explain what role is played by Correlation and Covariance in EDA?

Exploratory Data Analysis (EDA) involves understanding relationships between variables. Two key measures used are **Covariance** and **Correlation**.

Covariance

- **Definition:** Measures how two variables vary together.
- **Role in EDA:**
 - Identifies whether variables move in the **same direction (positive covariance)** or **opposite direction (negative covariance)**.
 - Helps detect **linear relationships** between variables.
- **Limitation:** Scale-dependent, making comparisons difficult.
- **Example:** In sales data, covariance between advertising spend and revenue shows if both increase together.

Correlation

- **Definition:** Standardized measure of the **strength and direction** of the linear relationship between variables.
- **Role in EDA:**

- Provides a **normalized value between -1 and +1**, making comparisons easier.
- Detects **strong, weak, or no relationship** between variables.
- Used to identify **multicollinearity** in datasets before modeling.
- **Example:** Correlation between height and weight shows how strongly they are related.

d. What are heuristic functions? Where are they used?

Definition:

A **heuristic function** $h(n)$ is an estimate of the cost or distance from a given state n to the goal state. It guides search algorithms by providing an *informed guess* about which path is more promising.

Key Points:

- **Purpose:** Improves efficiency of search by reducing exploration of unnecessary states.
- **Property:** Should be *admissible* (never overestimates the true cost).
- **Nature:** Problem-specific, designed to capture domain knowledge.
- **Notation:** Often used in algorithms like A^* , where $f(n) = g(n) + h(n)$.

Where They Are Used:

1. Informed Search Algorithms:

- A^* Search, Greedy Best-First Search.

2. Games:

- Evaluating board positions in Chess or Tic-Tac-Toe.

3. Pathfinding Problems:

- Navigation systems, robotics (e.g., Manhattan distance in grid maps).

4. Optimization Problems:

- Puzzle-solving (8-puzzle, 15-puzzle).

Examples of Heuristic Functions:

- **Misplaced Tiles (h1):** Counts tiles not in their correct position (8-puzzle).
- **Manhattan Distance (h2):** Sum of row + column distances each tile is away from its goal position.

e. What is Unification? Give example.

Definition:

Unification is a process in **predicate logic and AI** that makes two logical expressions **identical** by finding a suitable substitution of variables. It is a fundamental operation in **automated reasoning, inference engines, and Prolog programming**.

Key Points:

- Finds a **substitution mapping** for variables.
- Ensures two expressions can be matched for inference.
- Used in **resolution proofs** and **rule-based systems**.
- Essential for logical deduction in AI.

Example:

- Expressions:
 1. Loves(x,y)
 2. Loves(John,y)
- **Unification:** Substitution $x=John$.
- Result: Both become Loves(John,y).

Q2. [10 Marks each] - Answers

a. Write a detailed note on hypothesis testing. What are type I and type II errors?

Definition

Hypothesis testing is a **statistical method** used to make decisions or inferences about a population based on sample data. It helps determine whether the observed results are due to chance or represent a real effect.

Steps in Hypothesis Testing

1. Formulate Hypotheses

- **Null Hypothesis (H_0):** Assumes no effect or no difference.
- **Alternative Hypothesis (H_1):** Assumes there is an effect or difference.

2. Select Significance Level (α)

- Common values: 0.05 or 0.01.
- Represents the probability of rejecting H_0 when it is true.

3. Choose Test Statistic

- Examples: Z-test, T-test, Chi-square, ANOVA.
- Depends on data type and sample size.

4. Compute Test Statistic

- Calculate value from sample data.

5. Decision Rule

- Compare test statistic with critical value or p-value.
- If $p\text{-value} < \alpha$, reject H_0 .

6. Conclusion

- State whether evidence supports the null or alternative hypothesis.

Type I and Type II Errors

• Type I Error (False Positive):

- Rejecting H_0 when it is actually true.
- Probability of Type I error = α .

- Example: Concluding a drug works when it actually doesn't.
- **Type II Error (False Negative):**
 - Failing to reject H_0 when H_1 is true.
 - Probability of Type II error = β .
 - Example: Concluding a drug doesn't work when it actually does.

b. The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is a criminal!

Step 1: Represent Facts in Predicate Logic

Let:

- $American(x) \rightarrow x$ is American
- $Weapon(x) \rightarrow x$ is a weapon
- $Sells(x, y, z) \rightarrow x$ sells y to z
- $Hostile(x) \rightarrow x$ is hostile
- $Criminal(x) \rightarrow x$ is a criminal
- $Missile(x) \rightarrow x$ is a missile
- $Owns(x, y) \rightarrow x$ owns y
- $Enemy(x, America) \rightarrow x$ is enemy of America

Given Statements

1. Law:

- It is a crime for an American to sell weapons to hostile nations
- $\forall x \forall y \forall z$
- $American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \rightarrow Criminal(x)$

2. Nono is enemy of America:

- $\text{Enemy}(\text{Nono}, \text{America})$

3. Enemies are hostile:

- $\forall x$
- $\text{Enemy}(x, \text{America}) \rightarrow \text{Hostile}(x)$

4. Nono has missiles:

- $\exists x$
- $\text{Missile}(x) \wedge \text{Owns}(\text{Nono}, x)$

5. Missiles are weapons:

- $\forall x$
- $\text{Missile}(x) \rightarrow \text{Weapon}(x)$

6. All missiles of Nono were sold by West:

- $\forall x$
- $\text{Missile}(x) \wedge \text{Owns}(\text{Nono}, x) \rightarrow \text{Sells}(\text{West}, x, \text{Nono})$

7. West is American:

- $\text{American}(\text{West})$

Step 2: Logical Inference

From (2) and (3):

- $\text{Enemy}(\text{Nono}, \text{America}) \rightarrow \text{Hostile}(\text{Nono})$
 $\Rightarrow \mathbf{\text{Hostile}(\text{Nono})}$

From (4):

- There exists a missile m such that:
 - $\text{Missile}(m)$
 - $\text{Owns}(\text{Nono}, m)$

From (5):

- $\text{Missile}(m) \rightarrow \text{Weapon}(m)$
 $\Rightarrow \mathbf{\text{Weapon}(m)}$

From (6):

- $\text{Missile}(m) \wedge \text{Owns}(\text{Nono}, m) \rightarrow \text{Sells}(\text{West}, m, \text{Nono})$
 $\Rightarrow \mathbf{\text{Sells}(\text{West}, m, \text{Nono})}$

From (7):

$\Rightarrow \mathbf{\text{American}(\text{West})}$

Step 3: Apply Law

Using Law (1):

- $\text{American}(\text{West})$
- $\text{Weapon}(m)$
- $\text{Sells}(\text{West}, m, \text{Nono})$
- $\text{Hostile}(\text{Nono})$

$\Rightarrow \mathbf{\text{Criminal}(\text{West})}$

Q3. [10 Marks each] - Answers

a. Explain uniform cost search and best first search in detail with examples and compare.

Definition

Uniform Cost Search (UCS) and **Best First Search (BFS - heuristic based)** are search algorithms used to find paths in a state space. UCS is **cost-based (uninformed)**, while Best First Search is **heuristic-based (informed)**.

1. Uniform Cost Search (UCS)

Explanation

- UCS expands the node with the **lowest path cost $g(n)$** from the start node.
- It uses a **priority queue** ordered by path cost.
- It guarantees the **optimal solution** when all costs are positive.

Working Steps

1. Start with initial node (cost = 0).
2. Insert into priority queue.
3. Remove node with **lowest cost**.
4. If goal reached → stop.
5. Expand node and update costs of neighbors.
6. Repeat until goal is found.

Example

- Graph with costs:
 $S \rightarrow A$ (1), $S \rightarrow B$ (4), $A \rightarrow G$ (5), $B \rightarrow G$ (1)
- Paths:
 - $S \rightarrow A \rightarrow G = 6$
 - $S \rightarrow B \rightarrow G = 5$
- UCS selects **$S \rightarrow B \rightarrow G$ (optimal path)**.

2. Best-First Search

Explanation

- Selects node with **lowest heuristic value $h(n)$** .
- Uses estimate of distance to goal.
- Does not consider actual path cost.

Working Steps

1. Start from initial node.
2. Use heuristic function $h(n)$.
3. Expand node with smallest $h(n)$.
4. Continue until goal is reached.

Example

- Suppose heuristic values:
 - $h(A)=6$, $h(B)=2$, $h(G)=0$
- From S → chooses B (lowest heuristic) → reaches goal.

Aspect	Uniform Cost Search	Best First Search
Type	Uninformed search	Informed search
Evaluation Function	($g(n)$) (path cost)	($h(n)$) (heuristic)
Goal	Minimum cost path	Fastest path to goal
Optimality	Yes	No (not guaranteed)
Completeness	Yes	Not always
Data Structure	Priority queue (cost-based)	Priority queue (heuristic-based)
Efficiency	Slower (explores more nodes)	Faster (guided search)
Example Use	Shortest path problems	Route finding with estimate

b. Explain various stages in the Data analytics lifecycle.

The **Data Analytics Lifecycle** is a structured process that guides how raw data is transformed into meaningful insights for decision-making.

1. Discovery

- Understand business objectives and define the problem.
- Identify stakeholders, scope, and resources.
- Example: A retail company wants to analyze customer churn.

2. Data Preparation

- Collect data from multiple sources (databases, APIs, files).
- Clean data (handle missing values, duplicates, inconsistencies).
- Integrate and transform into a usable format.
- Example: Combine sales records, customer demographics, and transaction logs.

3. Model Planning

- Select appropriate analytical techniques (statistics, ML algorithms).
- Define metrics for success (accuracy, precision, recall).
- Example: Choose logistic regression for churn prediction.

4. Model Building

- Apply chosen algorithms to training data.
- Build predictive or descriptive models.
- Example: Train a classification model to predict churn probability.

5. Communicate Results

- Interpret model outputs and visualize findings.
- Present insights using dashboards, charts, or reports.
- Example: Show churn risk scores for each customer segment.

6. Operationalize

- Deploy the model into production systems.
- Integrate with applications or decision-making workflows.
- Example: CRM system alerts managers when a customer is at high churn risk.

7. Monitor & Optimize

- Continuously track model performance.
- Update/retrain models when data changes (concept drift).
- Example: Retrain churn model every quarter with new customer data.

b. Describe PEAS and also write down the PEAS representations for Medical diagnosis system.

Definition:

PEAS stands for **Performance measure, Environment, Actuators, Sensors**. It is a framework used to describe the components of an **Intelligent Agent** in Artificial Intelligence.

- **Performance Measure:** Criteria to evaluate success of the agent.
- **Environment:** External conditions in which the agent operates.
- **Actuators:** Mechanisms through which the agent acts on the environment.
- **Sensors:** Tools to perceive and gather information from the environment.

PEAS Representation for a Medical Diagnosis System

Component	Description
Performance Measure	Accuracy of diagnosis, early detection of diseases, minimizing false positives/negatives, patient satisfaction.
Environment	Hospitals, clinics, patient records, medical databases, symptoms, lab reports.
Actuators	Display system (diagnosis report), alerts to doctors, treatment recommendations, prescriptions.
Sensors	Patient data input (symptoms, medical history), lab test results, imaging scans (X-ray, MRI), wearable health devices.

Example

- A patient enters symptoms into the system.

- **Sensors** collect data (symptoms, test results).
- The system analyzes using AI models.
- **Actuators** output diagnosis and suggest treatment.
- **Performance Measure** is judged by accuracy, speed, and reliability of diagnosis.

Q6. [10 Marks each] - Answers

a. What are the different planning techniques? Explain with example.

Definition:

Planning in AI refers to the process of generating a sequence of actions that an agent must perform to achieve a specific goal. Different techniques are used depending on complexity, uncertainty, and flexibility of the environment.

1. State-Space Planning

- **Concept:** Represents problems as states and transitions.
- **Example:** Pathfinding in a maze where each position is a state.

2. Forward Planning (Progression Search)

- **Concept:** Starts from the initial state and applies actions step by step until the goal is reached.
- **Example:** Robot moving from warehouse → loading package → driving → delivering.

3. Backward Planning (Regression Search)

- **Concept:** Starts from the goal state and works backward to the initial state.
- **Example:** Solving a puzzle by imagining the final arrangement first.

4. Hierarchical Planning

- **Concept:** Breaks down complex goals into sub-goals using abstraction levels.

- **Example:** Planning a trip → Book tickets → Arrange accommodation → Plan sightseeing.

5. Partial-Order Planning

- **Concept:** Plans actions without fixing their exact order initially, allowing flexibility.
- **Example:** Cooking dinner → chopping vegetables and boiling water can be parallel tasks.

6. Conditional Planning

- **Concept:** Considers uncertainty and alternative paths depending on conditions.
- **Example:** Delivery agent → "If traffic jam → take alternate route; else → follow main road."